Talk Helps, but Excess Talk Can Hurt: Pre-vote Communication and the Swing Voter's Curse

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Abstract

This paper investigates the effect of pre-vote communication on collective decision-making when abstention is allowed. We present a model of majority-rule voting with common-interest and asymmetrically informed voters, with and without a communication stage. Communication does not alter the set of equilibrium outcomes. Still, we hypothesize that communication decreases efficiency, based on the conjecture that messages from poorly informed voters dilute those from better-informed voters. Our experimental implementation of the model shows that communication instead improves the efficiency of collective decisions, primarily through an unanticipated mechanism: it curbs irrational voting by less-informed voters. Nevertheless, consistent with our message-dilution conjecture, efficiency is lower when everyone speaks than when the less-informed voters stay silent, because after communication less-informed voters rarely abstain and instead largely vote with the majority message.

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1 Introduction

From judicial courts to academic committees, collective decisions made through majority voting are ubiquitous. Voting typically aggregates individuals' private information and is often preceded by some form of public communication among group members. Does pre-vote communication improve collective decisions, or can it be harmful? Over the past two decades, several studies have explored this issue. From theoretical work (Coughlan, 2000; Gerardi and Yariv, 2007; Austen-Smith and Feddersen, 2006) to experimental studies (Guarnaschelli et al., 2000; Goeree and Yariv, 2011), a central insight is that pre-vote communication can improve information aggregation by facilitating the revelation of private information, particularly when voters' preferences are sufficiently aligned. Yet this positive effect is not universal: Iaryczower et al. (2018), in a study of U.S. appellate court decisions, find that deliberation can be detrimental when judges have relatively aligned preferences and relatively precise private information.

Two important aspects remain underexplored. First, the mechanisms underlying detrimental communication remain unclear. While the concept of communication equilibria (Forges, 1986; Gerardi and Yariv, 2007) allows for the possibility of harmful equilibria in the voting game, and structural estimation from field data (Iaryczower et al., 2018) provides evidence of detrimental pre-vote communication, neither explains how communication leads to harm. Second, abstention has been abstracted away, despite its common occurrence in real-world situations. Abstention may provide a previously unexplored channel through which communication can harm information aggregation, as communication may alter voters' knowledge and strategic considerations when deciding whether to abstain.

This paper investigates how pre-vote communication affects collective decisions in environments where abstention is allowed. In the absence of communication, theory predicts, and several experiments find, that poorly informed voters abstain strategically (Feddersen and Pesendorfer, 1996, 1999; Battaglini et al., 2010; Morton and Tyran, 2011). The rationale, known as the *swing voter's curse* (Feddersen and Pesendorfer, 1996), is that if a poorly informed voter's vote is pivotal, his or her vote may cancel out that of a better-informed voter and push the outcome in the wrong direction. Therefore, it is rational for the voter to abstain and delegate the decision to better-informed voters, even when voting is costless. When communication is available, natural questions arise: Does strategic silence emerge analogously to strategic abstention? Does the presence of a communication stage change the extent of delegation to better-informed voters and, consequently, the

efficiency of decision-making? These are the central questions explored in this paper.

We study a voting model with common-interest and asymmetrically informed voters. Three voters make a binary group decision by majority rule. One *expert* receives a more accurate private signal than the two *non-experts*, with each voter privately observing his or her own signal and signal precision. We focus on the case where the expert's signal is accurate enough that deferring to the expert outperforms equal-weight aggregation. We compare three institutions: *no communication*, where voters vote immediately after receiving their signals; *voluntary communication*, where voters can choose to remain silent or send a message before voting; and *forced communication*, identical to voluntary communication except that silence is not allowed. Abstention from voting is allowed throughout. In both communication games, messages are restricted: voters can only send binary messages about the state, similar to the nonbinding straw poll of Coughlan (2000), so signal precision cannot be communicated. This captures committee procedures in which members communicate coarse recommendations or assessments without readily conveying the reliability of the underlying information.[1] Importantly, this protocol makes silence the communication-stage analogue of strategic abstention in the standard swing voter's curse. Given fully aligned preferences and majority rule, the effect of communication does not arise from strategic information transmission or strategic voting, but from its potential influence on whether voters effectively delegate to the expert.

Our model shows that all three games have multiple equilibria and share the same set of equilibrium outcomes. Hence, the impact of communication depends on equilibrium selection and is ultimately an empirical question. We hypothesize that both communication games yield lower group decision accuracy than the no-communication game, based on two behavioral conjectures. First, non-experts are not likely to abstain after a communication stage, because others' messages make them feel better informed and thus inclined to vote. Second, non-experts are inclined to speak even if they would abstain without communication, since speaking has no direct payoff consequences and sharing information can appear weakly beneficial in a common-interest setting. Together, we expect low-quality messages to dilute high-quality ones during communication, which then carries over to voting and results in less efficient group decisions.

We conducted a laboratory experiment implementing the three games. Contrary to our hypothesis, the availability of communication *improves* group decision accuracy,

[1] For example, some academic committees use secret-ballot straw votes before a final vote, and some use structured anonymous questionnaires that collect members' assessments and report the group's responses in aggregate, in the spirit of Delphi-style procedures. Such aggregate disclosures typically neither reveal differences in information quality underlying individual judgments nor weight judgments by that quality.

whether or not silence is allowed. However, when communication is available, accuracy is lower when everyone speaks than when only one voter speaks, supporting our message-dilution conjecture. In short, communication helps, but not everyone should speak.

Communication improves group accuracy primarily through an unanticipated mechanism: it curbs irrational voting by non-experts. Without communication, a non-negligible share of non-experts vote against their signals, substantially reducing group accuracy. We identify the gambler's fallacy as a source: non-experts' votes are significantly influenced by the previous round's true state despite the independence of states across rounds. Communication substantially attenuates this bias, shifting the votes of non-experts away from irrelevant past outcomes and toward information shared by other members. Consequently, group decisions aggregate private information more efficiently.

As a secondary mechanism, voluntary communication increases the frequency of equilibria in which decision-making is effectively delegated to the expert. Without communication, a single non-expert who does not abstain can render group decisions inefficient. Under voluntary communication, however, if one non-expert speaks while the other remains silent, the speaking non-expert can infer that the only other speaker is the expert and thus be persuaded; under majority rule, the group's decision effectively follows the expert.

We find support for our conjecture that communication discourages non-experts from abstaining. In both communication games, non-experts almost always vote. This pattern helps explain the finding that "excessive talk harms": when all voters speak, non-experts cannot assess the quality of others' messages and therefore tend to vote according to the majority message, potentially overwhelming the expert and reducing group accuracy. We do not, however, find support for our conjecture that strategic silence would be less frequent than strategic abstention: non-experts remain silent about as often as they abstain in the absence of communication, with both behaviors occurring in fewer than half of cases.

Our findings have implications for collective decision-making. First, pre-vote communication can be valuable, particularly in committees that repeatedly evaluate similar but largely independent cases, such as grant proposals or promotion files. Field evidence documents patterns consistent with the gambler's fallacy among individual asylum judges and in R&D funding panels (Chen et al., 2016a; Criscuolo et al., 2021). Our results show that voting itself can carry such history-based biases into group decisions, whereas pre-vote communication can substantially attenuate their influence. Second, when information quality varies substantially across members and is hard to convey, everyone speaking can

dilute high-quality messages and limit the benefit of communication. Norms or protocols that encourage poorly informed members to remain silent might reduce such dilution and help groups defer to better-informed members.

The paper is organized as follows. Section 2 reviews the related literature. Section 3 presents the model and hypotheses. Section 4 explains the experimental design and procedures. Section 5 reports the results. Section 6 discusses irrational votes and revisits evidence on strategic abstention. Section 7 concludes. Proofs and supplementary materials are provided in the Appendix.

2 Related literature

This paper relates to two main strands of literature. The first concerns the swing voter's curse. Feddersen and Pesendorfer (1996) show that it can be rational for poorly informed voters to abstain even when voting is costless. Battaglini et al. (2010) provide laboratory support for this prediction in settings where voters are either fully informed or completely uninformed. Morton and Tyran (2011) extend the framework by letting all voters receive informative but noisy signals, and find that less-informed voters largely abstain when the precision gap is large. Later experiments, however, document substantial uninformed voting in a variety of environments (Grosser and Seebauer, 2016; Buechel and Mechtenberg, 2019; Elbittar et al., 2020). Section 6.2 discusses this mixed experimental evidence together with our finding from the no-communication treatment.

In this literature, the role of pre-vote communication has received little attention. The closest study is that of Buechel and Mechtenberg (2019), who allow communication over networks rather than public communication among all group members. We contribute by analyzing the effect of pre-vote public communication, a relevant case for committee decisions, on the swing voter's curse. Specifically, we extend Morton and Tyran (2011) by adding a communication stage.

The second strand examines how pre-vote communication affects information aggregation in voting. Coughlan (2000) initiates this line of research by introducing a nonbinding straw poll before voting and showing that voters fully reveal their information if and only if preferences are sufficiently aligned. Gerardi and Yariv (2007) analyze free-form communication using the concept of communication equilibrium (Forges, 1986; Myerson, 1994), demonstrating that communication renders a wide range of voting rules equivalent in terms of sequential-equilibrium outcomes. Experimental work supports these effects. Guarnaschelli et al. (2000) find, in a common-interest setup, that subjects largely reveal

their private signals in a nonbinding straw poll and that the straw poll significantly improves information aggregation and efficiency by largely eliminating strategic voting under unanimity rule. Goeree and Yariv (2011) find evidence that unrestricted communication narrows differences between voting rules and improves efficiency. Notably, these studies abstract from abstention, focusing instead on how communication affects strategic voting. In contrast, this paper focuses on strategic abstention, a previously overlooked channel through which communication can affect information aggregation.

Field investigations of pre-vote communication are rare, primarily because key parameters—such as preferences, private signals, and the true state—are difficult to identify.[2] An exception is Iaryczower et al. (2018), who examine the impact of deliberation on U.S. appellate court decisions by estimating a free-form communication model based on Gerardi and Yariv (2007). They find that deliberation improves outcomes when judges are predisposed to disagree or have imprecise private information, but can worsen them otherwise. However, limitations in field data and the communication equilibrium framework leave the mechanisms behind these negative effects unclear. The present paper uses controlled experiments and incorporates abstention, offering a new avenue to identify when and how communication may harm voting outcomes.

Two further strands of literature are related to our study but differ in focus. First, our evidence on the gambler's fallacy in voting speaks to experimental work on bounded rationality and team decision making. Unlike our paper, that literature focuses on comparing group with individual decisions and, importantly, does not involve the aggregation of structured private information within groups. Studies generally find that groups are less susceptible to cognitive biases than individuals (for a survey, see Charness and Sutter, 2012). In particular, Baillon et al. (2016) separate the aggregation effect of a voting rule from the communication effect of group discussion, finding that the latter always improves rationality and the former only sometimes. We compare institutions with the group setting held fixed and therefore have no individual-decision benchmark. That said, our results echo this point; see Section 6.1.

Second, although our setting features heterogeneous information quality, it differs from work on competence signaling in committees (Levy, 2007; Visser and Swank, 2007; Fehrler and Hughes, 2018). That strand presumes reputational concerns—members care about being seen as well informed—and centers on how much expertise can be credibly transmitted and with what consequences. We assume no reputational payoff, following

[2]For further discussion of the challenges in field research and the advantages of experiments in this context, see Goeree and Yariv (2011).

the two literatures we build on above.

3 The model

3.1 Setup

Three voters make a group decision between options A and B. There are two possible states of the world, A and B, and the probability that state A occurs is $\pi = 0.5$, which is common knowledge. Each voter simultaneously chooses to vote for option A, vote for option B, or abstain. The group decision is the option that receives a majority of votes; ties are broken randomly. All voters have identical preferences, normalized so that each voter receives utility 1 if the group decision matches the true state and 0 otherwise.

Before voting, each voter i receives a private signal $s_i \in \{A, B\}$ about the true state of the world. There is one expert (labeled H) who receives a high-quality signal and two non-experts (labeled L_1 and L_2) who receive low-quality signals. This setup reflects real-world scenarios where well-informed individuals are outnumbered by a less-informed majority, creating tension between pooling information and deferring to the expert. Signals are independent across voters, conditional on the true state. The expert's signal matches the true state with probability q_H, i.e., $\Pr\{s_H = A \mid \omega = A\} = \Pr\{s_H = B \mid \omega = B\} = q_H$. A non-expert's signal matches the true state with probability q_L, i.e., $\Pr\{s_i = A \mid \omega = A\} = \Pr\{s_i = B \mid \omega = B\} = q_L$ for $i \in \{L_1, L_2\}$, with $1 > q_H > q_L > 0.5$. Each voter knows q_H, q_L, that the group contains one expert and two non-experts, and his or her own signal quality.

We analyze three voting games that differ in the availability and form of pre-voting communication. In the *No Communication (NC)* game, voters vote immediately after receiving private signals. In the *Voluntary Communication (VC)* game, a communication stage occurs after voters receive their signals but before voting. Each voter chooses to send one of three messages—"A," "B," or an empty message ($\varnothing$)—to all voters in the group. Note that voters only observe messages but not the quality of the underlying signals. After communication, voting proceeds as in NC. For convenience, we say a voter *speaks* if he or she sends a non-empty message, *speaks truthfully* if the message aligns with his or her signal, and *stays silent* if he or she sends an empty message. In the *Forced Communication (FC)* game, the rules are the same as in VC, except that voters must send either "A" or "B" before voting—empty messages are not allowed. We refer to VC and FC as communication games.

We examine pure-strategy perfect Bayesian equilibria (PBE). To sharpen the equilibrium analysis, we impose several assumptions on strategies. First, following the voting literature and our balanced-prior setup, we assume that voters respond to information if they choose to vote. Specifically, if all the signals and messages a voter observes were reversed, the voter would also reverse his or her vote. This assumption rules out trivial equilibria in which some voters always vote for the same option regardless of the information they receive. Second, we assume that voters speak truthfully if they choose to speak. This is natural given the common-interest structure and is supported by our experimental data: over 95% of non-empty messages are truthful in each communication game. Third, we assume that the expert never stays silent or abstains, similar to the assumptions in Fehrler and Hughes (2018) and Morton and Tyran (2011). Given the superior quality of the expert's signal, there is no reason to expect the expert to withhold information. Our experimental data support this assumption, showing that experts are silent or abstain in fewer than 3% of cases. These assumptions are formally defined in Appendix A.

We define an *outcome* as a mapping from the realizations of the expert's and non-experts' signals to (the probabilities of) group decisions. An *equilibrium outcome* is an outcome resulting from a PBE. [3] An outcome is *majority-decide (MD)* if, for every realization of signals, the group decision matches the signal received by most voters. An outcome is *expert-decide (ED)* if, for every realization of signals, the group decision matches the signal received by the expert. As will be shown later, there are no other equilibrium outcomes in any of the three games. To examine the performance of equilibria, we define *efficiency* as the ex-ante probability of having a group decision that matches the true state of the world. An outcome is more efficient if it chooses the correct group decision with a higher probability. Note that given the common-interest setup of our games, this variable is proportional to total expected payoff and reflects Pareto optimality.

Proposition 1 *The expert-decide (ED) and majority-decide (MD) outcomes yield efficiency* q_H *and* $2q_Hq_L(1-q_L)+q_L^2$*, respectively. ED is more efficient than MD if and only if* $q_H > q_L^2/(1-2q_L(1-q_L)) \equiv f(q_L)$.

The efficiency ranking reflects a trade-off between aggregating more pieces of information and equally weighting signals of unequal accuracy: as q_H and q_L diverge, delegating

[3] We do not specify outcomes for each state because, given an equilibrium profile and a specific signal realization, the group decision does not depend on the state. Additionally, we do not distinguish between the specific actions taken by voters along the equilibrium path. In other words, different equilibria that generate distinct paths are treated as yielding the same equilibrium outcome as long as they map signals to group decisions in the same way. These simplifications allow us to focus on how equilibria perform in terms of group decisions, making the three games comparable regarding their sets of equilibrium outcomes.

to the expert becomes more valuable compared to pooling all information and ED's advantage over MD grows. When $q_H > f(q_L)$, the expert is sufficiently better informed that *ED* dominates; when $q_H < f(q_L)$, voters' signal precisions are close and thus *MD* dominates. The efficiency gap is asymmetric across the two regions: ED's advantage approaches 25 percentage points as $q_H \to 1$ and $q_L \to 0.5$ (efficiency near 1 versus 0.75), whereas MD's advantage approaches only 10 points as $q_H \to q_L \approx 0.79$ (0.88 versus 0.79); see Appendix A.

In the following, we restrict ourselves to the ED-superior region $q_H > f(q_L)$, where the expertise asymmetry is relatively large and the efficiency gap can be sizable.

3.2 Equilibrium

We now characterize the equilibria of the three games, using the following terminology for strategy profiles. For the communication stage, a profile is *Hspeak* if the expert speaks truthfully while both non-experts remain silent, *Allspeak* if everyone speaks truthfully, and *H1Lspeak* if the expert and one non-expert speak truthfully while the other stays silent. For the voting stage, a profile is *Allvote* if all voters vote according to their posterior beliefs, *Hvote* if only the expert does so while both non-experts abstain, and *H1Lvote* if the expert and one non-expert vote according to their posterior beliefs while the other abstains. Profiles that differ only by relabeling the two non-experts are counted as one profile.

Proposition 2 *In each game, every equilibrium yields either the ED or the MD outcome, and both are equilibrium outcomes. Specifically:*

(i) The NC game has two equilibria: Allvote (MD) and Hvote (ED).

(ii) The VC game has eight equilibria: three Hspeak and three H1Lspeak equilibria, all yielding ED, plus Allspeak–Allvote (MD) and Allspeak–Hvote (ED).

(iii) The FC game has two equilibria: Allspeak–Allvote (MD) and Allspeak–Hvote (ED).

In the NC game, both equilibria and the welfare comparison follow directly from Morton and Tyran (2011). *Allvote* is an equilibrium because the third vote matters only when the other two disagree, in which case breaking the tie with private information beats abstaining. *Hvote* is an equilibrium in the spirit of the swing voter's curse: given that one non-expert abstains and the expert votes, the other non-expert is better off abstaining as well, because conditional on being pivotal his or her vote would counter the expert's

better-informed vote and harm the group decision. *Hvote* is more efficient than *Allvote* by Proposition 1 and our restriction $q_H > f(q_L)$. Note that *H1Lvote*, in which the expert and one non-expert vote while the other abstains, is not an equilibrium in this game: the voting non-expert can improve his or her payoff by abstaining, as the vote will either be inconsequential or cancel out the better-informed vote.

In the VC game, the set of equilibrium outcomes is unchanged from the NC game. Intuitively, adding a voluntary communication stage in this setup creates no new outcomes: the expert never switches the option he or she deems more likely, so communication either leaves the non-experts' posteriors unchanged or persuades one or both to follow the expert's signal, and the group decision remains determined by either the majority signal or the expert's signal. It also eliminates neither outcome.

The VC game is nonetheless richer than NC; a full characterization is in the proof of Proposition 2 in Appendix A. As the expert always speaks by assumption, equilibria can be organized by how many non-experts speak. Under *Hspeak*, both non-experts recognize that the sole speaker is the expert and are convinced by the spoken signal; regardless of whether they then vote or abstain, each voting profile constitutes an equilibrium after *Hspeak* and the decision is effectively made by the expert's signal. *Hspeak* equilibria thus shift the *Hvote* logic of NC from the voting stage to the communication stage: the non-experts' deference works through silence rather than abstention. Under *Allspeak*, the expert can no longer signal the superior quality of his or her information and non-experts follow the majority message, so the voting stage becomes decisive: in *Allspeak–Allvote* they vote and the decision tracks the majority signal, yielding MD; in *Allspeak–Hvote* they abstain and defer to the expert, yielding ED. Moreover, *H1Lspeak* also constitutes equilibria: the speaking non-expert recognizes that the only other speaker is the expert and is persuaded, so the outcome is again ED. Notably, the analogous voting profile *H1Lvote* is not an equilibrium in NC.

The FC game is the VC game where silence is not allowed, so only the *Allspeak* profiles survive: everyone speaks truthfully, and the voting stage decides the outcome—*Allspeak–Hvote* (ED) if the non-experts abstain, *Allspeak–Allvote* (MD) if they vote.

3.3 Equilibrium selection and hypotheses

Proposition 2 shows that the three games share the same two equilibrium outcomes, ED and MD. Thus, theory alone does not rank the games by efficiency. Which outcome is realized depends on two aspects of non-experts' behavior: whether they abstain at

the voting stage and, in VC, whether they remain silent at the communication stage; both abstaining and voting can arise in equilibrium, as can remaining silent and speaking in VC. Which behavior prevails, and hence which equilibrium is selected, is ultimately an empirical question. We state a behavioral conjecture for each aspect and derive the implications for equilibrium selection and efficiency.

3.3.1 Two behavioral conjectures

We first conjecture that non-experts rarely abstain after a communication stage. A communication stage, regardless of how many non-experts speak, enables non-experts to learn about others' signals, leaving them better informed about the state. Intuitively, more information can make voters more inclined to vote—whether for strategic reasons (Feddersen and Pesendorfer, 1996) or decision-theoretic ones (Matsusaka, 1995). Moreover, communication may make the strategic rationale for abstention less salient: non-experts must process others' messages in addition to conditioning on the event that their vote is pivotal. We thus expect non-experts to abstain only rarely once communication is available.

Conjecture 1 *Non-experts rarely abstain in both VC and FC.*

We next turn to silence. Without communication, Morton and Tyran (2011) found high non-expert abstention when the asymmetry in information quality was large.[4] We conjecture the silence rate in VC to be lower than the abstention rate in NC. Even if non-experts manage to largely abstain without communication, they may be reluctant to withhold a message once communication is allowed: unlike voting, speaking carries no direct payoff consequence, and its consequences for the group are realized through the subsequent voting stage. If non-experts narrowly bracket the communication decision, they may focus on the immediate implication that speaking provides the group with more information—which can appear beneficial in a common-interest environment—without fully accounting for its consequences for subsequent voting and the group decision (Read et al., 1999). Refraining from speaking may thus be harder than refraining from voting in the absence of communication.

Conjecture 2 *Non-experts stay silent in VC less often than they abstain in NC.*

[4]Morton and Tyran (2011) used the same setup as our NC game. In their treatment with $q_H = 0.9$ and $q_L = 0.65$, 87.86% of non-experts abstained, and 87% of groups in the final ten periods coordinated on the *Hvote* equilibrium (which they referred to as the *LTED*—let-the-expert-decide—equilibrium).

3.3.2 Implications for equilibrium selection and efficiency

Under Conjecture 1, the *Hvote* profiles—which require both non-experts to abstain—become implausible in both communication games, leaving *Allspeak–Allvote* (MD) as the only plausible FC equilibrium, and *Hspeak–Allvote* (ED), *H1Lspeak–Allvote* (ED), and *Allspeak–Allvote* (MD) in VC. Given the ex ante symmetry of the two non-experts, however, we do not expect them to coordinate on the asymmetric *H1Lspeak* equilibria. We therefore focus on the two symmetric communication profiles, *Hspeak* and *Allspeak*, which also correspond to the two extremes of the message-dilution channel. When both non-experts remain silent (*Hspeak*), the expert can signal the quality of his or her information and the outcome is ED; when both non-experts speak (*Allspeak*), their low-quality messages dilute the expert's, and because non-experts vote rather than abstain (Conjecture 1), this dilution carries into the group decision, yielding MD.

Hypothesis 1 *Efficiency is higher when only the expert speaks in VC than when everyone speaks in VC, and also higher than in FC.*

We refer to this prediction as the message-dilution hypothesis. Note that it relies crucially on our conjecture that non-experts rarely abstain after communication (Conjecture 1): if non-experts abstained after speaking, as in the *Allspeak–Hvote* equilibrium, the dilution of the expert's message would not carry into the group decision and efficiency would be restored.

We next compare efficiency across the communication and no-communication games. Under Conjecture 2, groups align with *Hspeak–Allvote* (ED) in VC less frequently than with *Hvote* (ED) in NC. Combined with the FC game's single plausible equilibrium *Allspeak–Allvote* (MD), this yields an ordering of equilibrium selection across games (Hypothesis 2), and hence of efficiency (Hypothesis 3).

Hypothesis 2 *Group behavior aligns with ED equilibria most often in NC, less often in VC, and least often in FC. The opposite holds for MD equilibria.*

Hypothesis 3 *Efficiency is highest in NC, intermediate in VC, and lowest in FC.*

Hypothesis 3 predicts detrimental communication: voluntary communication reduces efficiency, and forced communication reduces it further. This prediction may appear at odds with intuition from the cheap talk literature (see Crawford and Sobel (1982); Crawford (1998)), which suggests that communication can be beneficial—particularly when preferences are aligned. However, our hypothesis stems from a key feature of our setup:

the importance of abstention. While communication may indeed facilitate information transmission among the common-interest voters, it can still harm collective outcomes by influencing how much information—and whose—is ultimately aggregated, precisely because voters have the option to abstain. This ordering prediction hinges on both conjectures: Conjecture 1 renders *Hvote* implausible in the communication games, while Conjecture 2 makes ED play less frequent in VC than in NC.

Finally, we note that all the hypotheses above rest implicitly on the assumption that voters are rational and coordinate on equilibria.

4 Experimental design and procedures

We implemented three treatments (NC, VC, FC), each replicating one of the three games described above. In every round, the computer randomly placed a prize in a *Blue* or *Red* box with equal probability. One subject was randomly assigned to receive a high-precision signal that was correct with probability 90% ($q_H = 0.9$), while the remaining two subjects received lower-precision signals that were correct with probability 60% ($q_L = 0.6$).[5] In the VC treatment, each subject chose one of three messages to send: "My information suggests *Red*," "My information suggests *Blue*," or "nothing." The "nothing" option was not available in the FC treatment. In all treatments, each subject earned 3 euros in a round if the group selected the box containing the prize and 0 otherwise.

At the end of each round, subjects saw a results page displaying votes, the group decision, and the true prize location. By clicking the *history* button, they could review past rounds: their own signal, all group members' votes, and all group members' messages (in the two communication treatments), along with the group decision and the true box.

We conducted 10 sessions with 288 subjects, each playing 20 rounds. Before the 20 experimental rounds, subjects answered comprehension questions about the instructions and completed one unpaid practice round. At the start of each session, subjects were randomly assigned to matching groups of six, which remained fixed throughout the session. Each matching group was assigned to a single treatment, with 16 matching groups per treatment. In each round, groups of three were randomly formed within each matching group. For each group, the true box was re-selected by the computer, and one of the three members was randomly re-assigned the high-precision signal. After the final round, subjects completed a short post-experimental questionnaire collecting demographic in-

[5]Subjects were informed of their own signal precision and that one group member had 90% accuracy while two had 60% accuracy, without labels such as "expert" or "non-expert."

formation, feedback, and responses to an additional question testing the key Bayesian comparison.[6]

Subjects received payoffs from two randomly selected rounds. Additionally, each subject received a 3.5-euro show-up fee. The experiment took place at CentERlab, Tilburg University, in October and November 2023. Subjects spent 20–35 minutes in the lab and earned an average of 8.68 euros.[7] Experimental instructions are provided in Appendix B.

5 Experimental results

First, in Section 5.1, we will compare group efficiency across treatments (Hypothesis 3) and across different numbers of speaking voters (Hypothesis 1). In Section 5.2, we will examine subjects' communication and voting behavior (Conjecture 1 and Conjecture 2). Finally, in Section 5.3, we will analyze equilibrium selection (Hypothesis 2).[8]

5.1 Group efficiency

In line with the model's definition, we measure group efficiency as the frequency with which groups choose the correct box. As a benchmark, given $q_H = 0.9$ and $q_L = 0.6$, the theoretical efficiency predictions are 79.2% for *majority-decide (MD)* equilibria and 90% for *expert-decide (ED)* equilibria (Proposition 1).

5.1.1 Efficiency across treatments

Table 1, column (1), presents the efficiency levels for the three treatments and the p-values of Mann–Whitney tests comparing each pair of treatments. The efficiencies of the two communication treatments are approximately 85%, between the predictions of the two equilibrium outcomes. In contrast, the NC treatment achieves 78.4% efficiency, marginally lower than the worst equilibrium outcome. Efficiency increases significantly in

[6]Specifically, subjects were asked which box the group should choose to maximize its winning probability when one member received a BLUE signal with a 90% correctness probability and two others received RED signals with a 60% correctness probability. About 78% of subjects selected BLUE, the Bayesian-optimal option, suggesting a reasonably good understanding of the key Bayesian comparison. The detailed questionnaire is provided in Appendix B.

[7]The experiment was programmed using oTree (Chen et al., 2016b). The design and the key hypothesis (Hypothesis 3) were pre-registered on AsPredicted (`https://aspredicted.org/wc55-vb5t.pdf`). A pilot was pre-registered separately (`https://aspredicted.org/27sx-sjk4.pdf`) whose data are not used here.

[8]Restricting the sample to the subjects who answered the post-experimental understanding question correctly leaves the individual-level results qualitatively unchanged. Group-level outcomes cannot be meaningfully tested on this subsample, as group efficiency is jointly determined by all three members and most groups retain only a subset of them.

the VC and FC treatments compared to NC, but there is no significant difference between the two communication treatments. These findings suggest that adding a communication stage improves efficiency, contradicting H3.

Table 1: Efficiency by treatment and difficulty of signal realizations

Treatment	Overall	By difficulty		
	(1)	Easy (2)	Medium (3)	Tricky (4)
NC	78.4%	88.6%	79.2%	56.0%
	(2.1)	(2.8)	(2.9)	(5.8)
VC	85.6%	95.2%	89.4%	59.2%
	(1.6)	(1.7)	(1.8)	(4.4)
FC	84.7%	95.0%	90.0%	55.5%
	(1.3)	(1.7)	(2.0)	(4.7)
# observations	1920	645	911	364
Mann–Whitney tests, two-sided, p-values				
NC vs VC	0.005***	0.037**	0.011**	0.596
NC vs FC	0.007***	0.048**	0.006***	0.806
VC vs FC	0.732	0.950	0.734	0.449
Theoretical predictions				
MD equilibrium	79.2%	95.3%	90.0%	20.0%
ED equilibrium	90.0%	95.3%	90.0%	80.0%

Notes: Each observation corresponds to a randomly re-matched group in a given round. Each treatment contributes 640 observations and 16 matching groups in total. Standard errors (in parentheses) are clustered at the matching-group level, and tests are computed on matching-group means. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

We further analyze efficiency differences by categorizing rounds based on the difficulty of making correct group decisions given the observed signals. A round is classified as *easy* if all three subjects in the group receive signals of the same color; *medium* if the expert and one non-expert receive the same signal, which contradicts the other non-expert; and *tricky* if the expert's signal contradicts the signals of both non-experts. Theoretical predictions for efficiency under majority-decide (MD) and expert-decide (ED) outcomes for these signal realizations are displayed at the bottom of Table 1.[9] In *easy* and *medium* cases, efficiency is predicted to be the same under MD and ED. However, in *tricky* cases, an MD outcome is expected to yield an efficiency of 20%, much lower than the 80% predicted under ED.

[9]The theoretical efficiency of an MD outcome (or an ED outcome) under different signal realizations is the probability that the majority signal (or the expert's signal) matches the true state, conditional on the realizations. For example, the theoretical efficiency of the ED outcome in *tricky* cases is, without loss of generality, $P(\omega = A \mid s_H = A,\ s_{L_1} = s_{L_2} = B) = (0.9 \cdot 0.4 \cdot 0.4)/(0.9 \cdot 0.4 \cdot 0.4 + 0.1 \cdot 0.6 \cdot 0.6) = 0.8$.

Table 1, columns (2)–(4), report experimental efficiency by signal-realization type and Mann–Whitney tests across treatments. While theory predicts that any efficiency differences across treatments would arise only in *tricky* cases, the laboratory results show a different pattern: communication improves efficiency in *easy* and *medium* cases but has no significant effect in the *tricky* case. Specifically, in *easy* rounds, groups in NC make correct decisions only 88.6% of the time, well below the 95% predicted under either equilibrium outcome, and significantly below the two communication treatments (both close to 95%). This pattern is striking because it implies that even when all group members receive a signal in favor of the red box, groups sometimes choose the blue box.[10] The same qualitative pattern, and similar concerns, arise in *medium* cases. We return to these efficiency patterns in Section 5.3, where we examine how equilibrium and non-equilibrium play contribute to them.

Result 1 *H3 is rejected. Efficiency is higher in VC and FC than in NC. This efficiency gain occurs in non-tricky rounds, that is, in rounds where the expert's signal is also received by at least one non-expert.*

5.1.2 Efficiency by the number of speaking voters

How does the amount of communication affect group efficiency? Table 2 reports efficiency for groups with varying numbers of speaking voters—voters who send non-empty messages—in the two communication treatments. Note that $n = 1, 2, 3$ speaking voters correspond to $0, 1, 2$ speaking non-experts, since the expert speaks in every round, as assumed in the model and confirmed by the data (see Section 5.2.1).

In the VC treatment, when only one voter speaks, group efficiency is 90.5%. When all three voters speak, efficiency drops to 83.1%, significantly lower than in the former case (Wilcoxon signed-rank test, $p = 0.042$). Efficiency when two voters speak lies between these two levels, but neither comparison involving two speakers is significant. In the FC treatment, where everyone speaks by design, group efficiency is 84.7%, also significantly lower than when only one voter speaks in VC (Mann–Whitney test, $p = 0.006$). The remaining comparisons between FC and the other VC cells are not significant.

Result 2 *H1 is confirmed. Efficiency is higher when only one voter speaks in VC than when all voters speak in VC, and it is also higher than in FC.*

[10] One alternative possibility is that groups chose the red box when receiving three red signals but were unlucky in that all three signals were incorrect. However, in the *easy* cases of NC, the true state matches participants' signals 96.2% of the time in the data, making this explanation unlikely.

Table 2: Efficiency by number of speaking voters in communication treatments

	VC			FC
	$n = 1$	$n = 2$	$n = 3$	$n = 3$
Efficiency	90.5%	85.4%	83.1%	84.7%
	(2.0)	(3.1)	(1.8)	(1.3)
# group-rounds	147	233	260	640
# matching groups	11	15	15	16

Notes: Each observation corresponds to a randomly re-matched group in a given round. n denotes the number of speaking voters, i.e., the number of non-empty messages in the group. Standard errors (in parentheses) are clustered at the matching-group level. Not every matching group produced rounds with each value of n, so the number of matching groups underlying each VC cell is reported separately. In the experiment, VC never had rounds with $n = 0$; FC always had $n = 3$ by design.

This finding highlights that excessive communication can be detrimental, as predicted by our message-dilution hypothesis (H1).[11] The prediction is not trivial: messages could in principle be harmless in a cheap-talk setting—if non-experts abstained after speaking, efficiency could be restored—so it hinges on the conjecture that non-experts rarely abstain in the communication games (Conjecture 1), which we examine below.

Two remarks are in order. First, despite Result 2, the treatment-level comparison in Table 1 shows no significant difference between VC and FC. The reconciliation is compositional. VC improves on FC only when groups restrain themselves to let the expert speak alone ($n = 1$), but they did so in only 23.0% of rounds, and the remaining rounds do not differ significantly in efficiency from FC. As a result, little aggregate difference remains between VC and FC. Second, because n is not randomly assigned within VC, the efficiency differences across n could in principle reflect selection rather than the effect of how many voters speak. The scope for such selection is nevertheless limited along important dimensions. The composition of signal precisions is fixed by design, and the number of speakers is not significantly associated with round difficulty.[12] The remaining concern is selection across subjects: non-experts who are more inclined to speak may also be poorer voters, in which case the lower efficiency at $n = 3$ could reflect poorer voting rather than message dilution. The analysis below of individual behavior and group coordination speaks against this interpretation, and we return to it at the end of Section 5.3.[13]

[11] This finding might loosely relate to an observation by Goeree and Yariv (2011) that correct decisions tend to be associated with shorter communication phases. However, this was not their primary focus, and the underlying mechanisms were not discussed.

[12] The design limits the scope for selection on round difficulty because silence decisions are made without observing others' signals. Consistent with this, a Cochran–Mantel–Haenszel test finds no significant association between the number of speakers and round difficulty ($p = 0.871$).

[13] Supplementary evidence also speaks against this: the share of non-experts answering the post-experimental understanding question correctly is 73.5%, 78.3%, and 81.9% in VC rounds with $n = 1$,

5.2 Individual behavior

5.2.1 Experts' behavior

In our equilibrium analysis, we assume that experts always speak truthfully in communication and always vote their signals in all three games. These assumptions are strongly supported by the experimental results. In the VC and FC treatments, experts never stay silent and they truthfully share their signals approximately 99% of the time. Across all three treatments, they vote according to their signals about 98% of the time. Specifically, in the two communication treatments, the expert votes according to his or her signal 99.64% of the time when at least one non-expert speaks in line with it, and 91% of the time when both non-experts' messages contradict it.

5.2.2 Non-experts in NC

In NC, non-experts abstain 44.5% of the time, vote according to private signals 40.2% of the time, and vote against private signals 15.4% of the time. Two aspects are noteworthy. First, the abstention rate is far lower than the 87.9% observed in the comparable treatment of Morton and Tyran (2011). In other words, our experiment does not reproduce Morton and Tyran's finding that, with a large information gap and without communication, less-informed voters largely delegate decision-making to the well-informed. We return to this discrepancy in Section 6.2. Second, non-experts vote contrary to their private signals in a non-negligible share of cases.[14] This is surprising because, in the NC game, subjects receive no additional information beyond their signals before voting, and voting against one's signal is never optimal. We explore the drivers of these irrational votes in Section 6.1.

5.2.3 Non-experts' communication

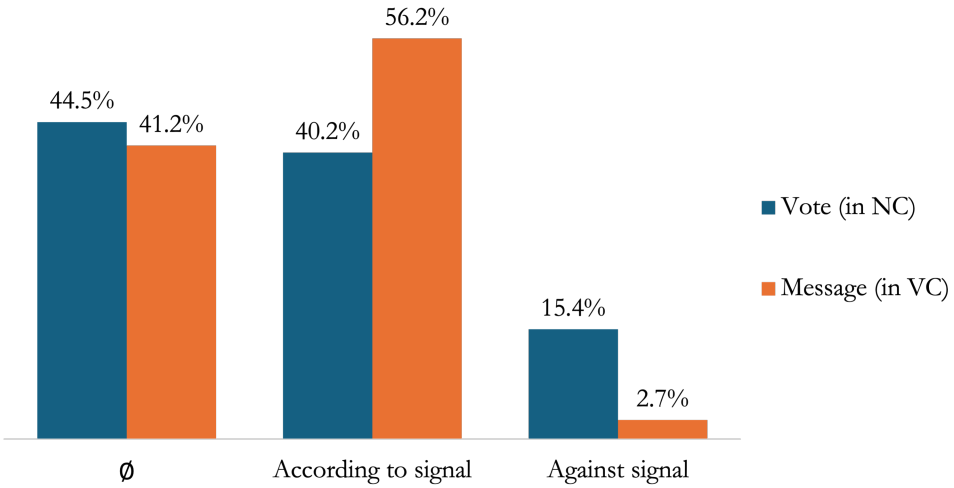
A key question in VC is whether non-experts remain silent, particularly relative to their tendency to abstain in NC. Figure 1 presents a comparison between non-experts' communication behavior in VC and their voting behavior in NC, the latter of which was discussed above.

As Figure 1 shows, non-experts in VC remain silent 41.2% of the time, which is not significantly different from their 44.5% abstention rate in NC. Thus, strategic silence occurs, but in fewer than half of cases—much like strategic abstention in NC. This finding

$n = 2$, and $n = 3$ respectively, so if anything, groups in which everyone speaks contain subjects who perform better on the understanding question.

[14]The rate of 15.4% exceeds typical error rates and those reported in similar studies, such as 5.9% in Buechel and Mechtenberg (2019) and 1.76% in Morton and Tyran (2011) for votes contrary to signals.

Figure 1: Non-experts voting behavior in NC and communication behavior in VC



Notes: Following the model, we use the empty symbol ∅ to denote abstention in NC and silence in VC.

contrasts with our prediction in Conjecture 2, which anticipated a lower rate of strategic silence than strategic abstention.

Result 3 *Conjecture 2 is not supported. Non-experts' abstention rate in NC is not significantly different from their silence rate in VC.*

Additionally, non-experts in VC send messages opposite to their signals only 2.7% of the time (and 4.5% conditional on speaking), supporting the model's assumption that once voters speak, they speak truthfully. In the FC treatment, non-experts speak truthfully 98.2% of the time, providing further support for this assumption.

5.2.4 Non-experts' voting after communication

How do non-experts behave in the voting stage following communication? Consistent with Conjecture 1, non-experts rarely abstain in the communication treatments, with abstention rates of 4.5% in VC and 9.5% in FC, compared with 44.5% in NC (pairwise chi-squared tests of proportions with Holm correction, both $p < 0.001$). This is the mechanism behind Result 2: because non-experts almost always vote after communication, the dilution from low-quality messages carries into the group decision rather than being undone by abstention. The abstention rate is also higher in FC than in VC ($p < 0.001$), although it remains uncommon in both treatments. One possible explanation is that FC removes the silence option, leaving abstention as the only form of non-participation and thus making the abstention decision more salient and the strategic reasoning behind it easier to attend to than in VC.

Result 4 *Conjecture 1 is supported. Abstention among non-experts is limited in both communication treatments.*

For non-experts who do not abstain, the equilibria predict that they update by Bayes' rule and vote according to their posterior; concretely, whenever a majority message exists—either because an odd number of voters speak or because two speakers send the same message—they should follow it. The data bear this out. Table 3 reports non-experts' voting behavior in the rounds where a majority message exists (91.6% of all rounds in the two communication treatments). When only one voter speaks, non-experts follow that message 97.6% of the time, consistent with their inferring that the lone speaker is the expert and being persuaded accordingly. When two voters speak—a majority message exists only if the two agree—non-experts follow it 93.2% of the time. When all three voters speak in VC, they follow the majority message 86.0% of the time: lower than in the previous cases, plausibly reflecting the increased difficulty of inference and coordination, but still high. In FC, where all voters speak by construction, the follow rate is 78.0%, with the remainder split between abstention (9.5%) and votes against the majority message (12.6%). Overall, non-experts who vote after communication generally follow the majority message when one exists. Detailed voting patterns for different message scenarios, including cases without a majority message, are reported and discussed in Appendix C.1.

Table 3: Non-expert voting after communication (when a majority message exists)

Vote follows the majority message?	VC			FC
	n=1	n=2	n=3	n=3
Follow	97.6%	93.2%	86.0%	78.0%
Abstain	2.4%	4.4%	2.5%	9.5%
Against	0.0%	2.4%	11.5%	12.6%
# observations	294	250	520	1280

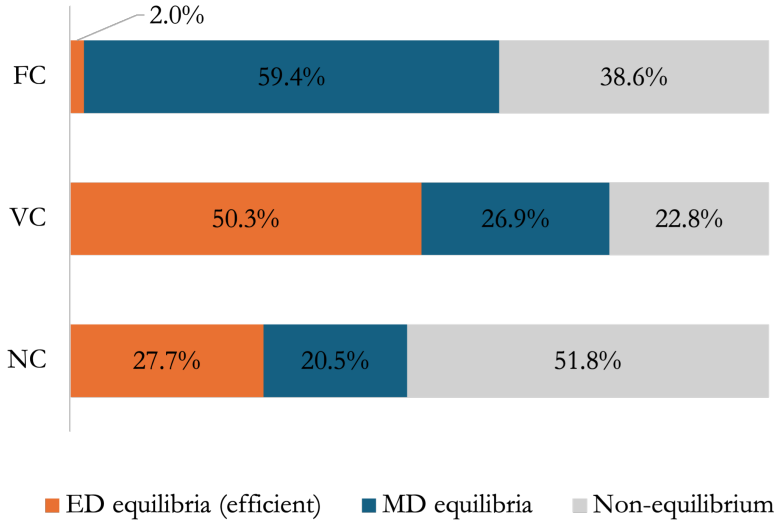
Notes: "Follow" refers to voting in line with the majority message, while "Against" refers to voting against it. n denotes the number of speaking voters. The $n = 2$ case includes only rounds in which the two voters send the same message, as there is no majority message otherwise.

5.3 Equilibrium selection

Does communication facilitate or hinder coordination on efficient equilibria? We say that a three-player group in a given round coordinates on an equilibrium if its observed behavior is consistent with some equilibrium path. Otherwise, we classify the group as

playing a non-equilibrium profile. Figure 2 shows, for each treatment, the frequencies of group coordination on ED and MD equilibria, as well as the frequency of non-equilibrium play. Table 4 reports the frequencies of groups coordinating on specific profiles, along with the efficiencies of the corresponding groups.

Figure 2: Frequencies of group coordination on equilibrium and non-equilibrium play



Three findings regarding coordination are noteworthy. First, only 48.2% of NC groups coordinate on an equilibrium, considerably fewer than in VC (77.2%, $p < 0.001$) and FC (61.4%, $p = 0.090$). As shown in Table 4, non-equilibrium play in NC primarily takes two forms: 23.3% of groups experience coordination failure between non-experts, where one abstains and the other does not, and 28.3% include at least one voter casting a vote against his or her private signal. These instances of non-equilibrium play largely account for the lower efficiency of the NC treatment: while groups playing *Hvote* or *Allvote* achieve efficiencies of 91.0% and 80.2%, respectively, groups with coordination failures among non-experts achieve only 74.5%, and those with irrational votes are correct just 68.5% of the time.

Second, groups in VC coordinate on ED equilibria in 50.3% of cases, significantly more than in NC (27.7%, $p = 0.045$) and in FC (2.0%, $p < 0.001$). As shown in Table 4, 23.0% of VC groups align with an equilibrium where only the expert speaks (*Hspeak*) and 27.3% with one where the expert and one non-expert speak (*H1Lspeak*). It is instructive to compare *H1Lspeak* equilibria in VC (one non-expert stays silent) with *H1Lvote* in NC (one non-expert abstains), as the non-experts behave asymmetrically in both. These profiles occur at similar rates (27.3% vs. 23.3%) but yield substantially different efficiencies (89.1% vs. 74.5%). As mentioned in Section 3, in VC, one non-expert speaking does not

Table 4: Frequency and efficiency of group coordination on each profile

Game	Profile	Frequency	Efficiency
NC	Hvote (ED)	27.7%	91.0%
	Allvote (MD)	20.5%	80.2%
	Non-equilibrium: H1Lvote	23.3%	74.5%
	Non-equilibrium: Irrational votes	28.3%	68.5%
	Non-equilibrium: Others	0.3%	50.0%
VC	Hspeak equilibria (ED)	23.0%	90.5%
	H1Lspeak equilibria (ED)	27.3%	89.1%
	Allspeak-Allvote (MD)	26.9%	83.1%
	Allspeak-Hvote (ED)	0.0%	-
	Non-equilibrium	22.8%	79.5%
FC	Allspeak-Hvote (ED)	2.0%	92.3%
	Allspeak-Allvote (MD)	59.4%	86.3%
	Non-equilibrium	38.6%	81.8%

Notes: In NC, "H1Lvote" denotes the non-equilibrium group behavior in which the expert and one non-expert vote according to their private signals, while the other abstains. "Irrational votes" denotes the non-equilibrium group behavior in which at least one voter votes against his or her private signal. Equilibrium profiles for each game follow the definitions in Section 3, with "MD" or "ED" in brackets indicating the corresponding equilibrium outcome.

reduce efficiency in an *H1Lspeak* equilibrium because the speaking non-expert can be persuaded by the expert's message.[15] In contrast, in NC, one non-expert voting leads to coordination failure and lower efficiency. The differing outcomes of these two profiles drive the higher rate of *ED* equilibrium play and the higher overall rate of equilibrium play in VC relative to NC, contributing to the efficiency gain observed in Section 5.1.[16]

Third, FC groups coordinate on *ED* equilibria much less and on *MD* equilibria much more than in the other two treatments (all $p < 0.001$), consistent with Hypothesis 2. Only 2.0% coordinate on *Allspeak-Hvote*, the only *ED* equilibrium in FC, consistent with the rare abstention observed after communication. This poor equilibrium selection might have been expected to depress FC's efficiency, yet Section 5.1 shows that FC outperforms NC and does not differ significantly from VC. The former indicates that non-equilibrium play in NC is the more costly of the two: NC groups reach an equilibrium only 48.2% of the time, and the profiles they otherwise fall into—coordination failure at 74.5% and irrational votes at 68.5%—are considerably less efficient than the *MD* equilibrium FC

[15]This persuasion pattern is also visible in voting behavior. When a non-expert speaks in line with his or her own signal but observes an opposing message from the only other speaker, he or she subsequently votes in line with that speaker's message 68.0% of the time. See Table C1 and Appendix C.1.

[16]On the other hand, the rate of *MD* equilibrium play does not differ significantly between VC (*Allspeak-Allvote*, 26.9%) and NC (*Allvote*, 20.5%), with $p = 0.258$.

mostly settles on.[17] The latter is compositional, as discussed following Result 2.

Result 5 *Contrary to H2, group behavior in VC aligns with ED equilibria more often than in NC. Consistent with H2, FC groups align with ED equilibria least often and with MD equilibria most often among the three treatments.*

As identified above, the two main sources of the efficiency gains from communication correspond to the two non-equilibrium profiles in NC that drive its low efficiency: irrational votes and coordination failure. This also clarifies why the gains arise only in non-tricky cases, as shown in Section 5.1. Irrational votes in NC can lead to incorrect group decisions in *easy* and *medium* rounds, where correct choices would otherwise be highly likely. In *tricky* rounds, however, they can sometimes improve outcomes by offsetting the inefficiency caused by *Allvote*—two wrongs make a right. On the other hand, the coordination failure (*H1Lvote* profile) in NC would not affect the efficiency of *easy* rounds, where all non-abstaining players would vote the same way, but it can reduce efficiency in *medium* and *tricky* rounds by canceling out the expert's vote with that of a non-expert. As a result of these two forces, NC underperforms VC and FC in *easy* and *medium* rounds. In *tricky* rounds, the finding that NC does not perform worse than the other two suggests that the two forces above offset each other. Detailed group efficiency by profile and round difficulty is reported in Table C2 in Appendix C.

We now return to the selection concern raised after Result 2: the lower efficiency at $n = 3$ could reflect poorer voting by those more inclined to speak rather than excess talk per se. Two observations limit this concern. First, as shown in Section 5.2.4, non-experts vote rationally most of the time in every n cell. Second, and more importantly, Table 4 indicates that the dilution effect is not driven by non-equilibrium play: restricting attention to VC groups whose behavior is consistent with an equilibrium path, efficiency still falls from 90.5% under *Hspeak* to 83.1% under *Allspeak-Allvote* (the only equilibrium observed under *Allspeak*), reproducing the raw efficiency gap between one and three speakers in Table 2.[18] Hence, the dilution effect is unlikely to be driven primarily by selection and instead reflects the predicted harm from having too many speakers under the *Allspeak-Allvote* equilibrium: because non-experts rarely abstain after communication and largely follow the majority message (Table 3), the two low-quality messages can outweigh the expert's message and reduce group efficiency.

[17] One may notice that FC groups playing the *MD* equilibrium achieve 86.3% efficiency, above the unconditional benchmark of 79.2%. This is because *tricky* rounds more often produce non-equilibrium profiles and are therefore under-represented among *MD*-classified rounds.

[18] Relatedly, once everyone speaks in VC, non-equilibrium voting causes no further efficiency loss: non-equilibrium groups achieve 83.0%, essentially identical to the 83.1% of *Allspeak-Allvote*.

6 Discussion

Two behavioral patterns among non-experts in NC stand out from Section 5: the non-negligible share of irrational votes and the abstention rate below one half. In this section, we investigate the source and implications of the former and place the latter in the context of the experimental literature on strategic abstention.

6.1 Gambler's fallacy in voting and the role of communication

One anomaly concerns the non-experts' votes in NC — they vote against their signals 15.4% of the time, which, as discussed above, is a primary source of NC's low efficiency. Voting against one's signal absent communication is irrational since it is strictly worse than both voting with it and abstaining, and therefore never arises in equilibrium (Lemma 1, in the Appendix). Given that the signal is the only information available at the time of voting, why do some non-experts vote against it? Below we identify the gambler's fallacy as a source of irrational voting, and show how communication attenuates it.

Table 5 presents a logistic regression of whether non-experts vote against their signals. Given the neutrality between *Red* and *Blue*, we assume a subject receives a signal for option A (either color), with the other option referred to as B. The dependent variable equals 1 if the voter votes against his or her signal and 0 otherwise. The variable *Past truth A* equals 1 if the previous round's true state was A. *Past abstain* and *Past vote against signal* equal 1 if the voter took the respective action in the previous round, *Past win* equals 1 if the voter won in the previous round, and their interactions capture reinforcement from winning. *Number of other message A* (respectively B) counts the messages for A (respectively B) sent by the other two voters; these variables are defined only for communication games and are included in columns (4) and (5).[19]

In NC (column (1)), the coefficient on *Past truth A* is positive and significant: a non-expert is more likely to vote against his or her signal when that signal matches the previous round's true box than when it does not. For example, after a *Red* true box, a non-expert is more likely to vote *Blue* in the next round despite receiving a *Red* signal. This is consistent with the gambler's fallacy—the mistaken belief in a negative correlation between independent trials (Tversky and Kahneman, 1971).[20] In NC, the only information available when voting is the private signal and the history of outcomes. A rational voter

[19]The robustness checks with more specifications, as well as regressions using the other two voting behaviors (vote for signal, abstention) can all be found in the appendix.

[20]Tversky and Kahneman (1971) attribute this fallacy to the representativeness heuristic: individuals expect small samples to reflect large-sample proportions and anticipate quick reversion to a 50:50 ratio.

Table 5: Logistic estimation of non-expert voting against signal (assume signal A)

Dependent variable: 1 if the non-expert votes for B (against signal A), 0 otherwise.					
	Baseline specification			With message variables	
	NC	VC	FC	VC	FC
	(1)	(2)	(3)	(4)	(5)
Past truth A	0.631***	0.030	0.044	0.097	0.320*
	(0.225)	(0.161)	(0.095)	(0.208)	(0.165)
Past abstain	-1.644***	-0.917	-1.411	0.692	-1.445*
	(0.586)	(1.230)	(1.243)	(5.015)	(0.872)
Past vote against signal	1.684***	-0.180	0.715**	0.310	0.808*
	(0.504)	(0.379)	(0.326)	(0.689)	(0.423)
Past win	-0.044	-0.152	-0.078	0.499	-0.257*
	(0.264)	(0.197)	(0.132)	(0.318)	(0.151)
Past win × Past abstain	-1.193	-0.250	0.453	-2.822	0.086
	(1.069)	(1.278)	(1.270)	(5.345)	(1.163)
Past win × Past vote against signal	0.729	0.178	-0.340	-0.210	0.100
	(0.545)	(0.403)	(0.271)	(0.696)	(0.286)
Number of other message A				-2.854***	-3.396***
				(0.425)	(0.255)
Number of other message B				1.535***	
				(0.336)	
Constant	-2.082***	-0.487**	-0.683***	-0.479	2.195***
	(0.323)	(0.202)	(0.132)	(0.373)	(0.316)
Log likelihood	-407.00	-781.35	-766.80	-360.27	-413.08
# observations	1,216	1,216	1,216	1,216	1,216

Notes: Standard errors, reported in parentheses, are adjusted for clustering at the matching-group level. Since communication is compulsory in FC, the number of messages for B equals two minus the number for A, so only the latter is included in column (5). $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

ignores the history of the true box, since it is independent across rounds. A voter prone to the gambler's fallacy, however, may expect a *Blue* true box after a *Red* one, assuming that under the 0.5 prior the two colors should balance out in the short run; since signals are noisy (60% accurate), such a non-expert may prioritize this heuristic over his or her signal. Experts, whose signals are far more precise (90% accurate), override such historical biases and rarely vote against their signals as found in the data. This effect of *Past truth A* in NC is robust across different specifications, as shown in Table C3 in the Appendix.

The effect of past truth, however, is much weakened in the two communication games. As shown in columns (2) and (3), *Past truth A* no longer significantly drives voting against one's signal in VC or FC. Formal tests show that the past-truth effect is significantly larger in NC than in VC ($p = 0.031$) and than in FC ($p = 0.018$), and not different between the two communication treatments ($p = 0.938$).[21] What drives the attenuation? As shown

[21] These tests come from a pooled specification in which the treatment indicator is interacted with every regressor of column (1), with standard errors clustered across all 48 matching groups. The specification

in Section 5, communication enables substantial information exchange—experts always speak, non-experts often do, and whoever speaks is almost always truthful. At voting time, subjects incorporate these messages alongside their signals and history. Even if prone to the gambler's fallacy, they are less likely to follow it since they now also receive information about others' signals, always including that of the expert. Columns (4) and (5) confirm that non-experts respond strongly to others' messages: the coefficients on *Number of other message A* and *Number of other message B* are statistically significant with the expected signs and sizable in magnitude. Communication thus shifts voting away from a heuristic based on the previous realized state and toward the information others share in the current round.

Result 6 *In NC, non-experts' voting is affected by the gambler's fallacy: a non-expert is more likely to vote against his or her signal when it agrees with the previous round's true box than when it does not. This effect is attenuated in VC and FC.*

Two remarks on this finding are in order. First, the gambler's fallacy is not a laboratory artifact, but a potentially relevant bias in legislative and committee settings in which similar but statistically independent cases are evaluated in sequence and previous outcomes may exert an influence they should not have. Field evidence exists in several high-stakes environments. For example, Chen et al. (2016a) document negatively autocorrelated decisions unrelated to case merits among asylum judges, loan officers, and baseball umpires, and Criscuolo et al. (2021) show that, in panel decisions, an R&D proposal reviewed immediately after a funded one receives a smaller share of the funding it requests.[22]

Second, we show that individual heuristics can survive into voting outcomes and reduce group efficiency, whereas even limited communication substantially attenuates their effect. Baillon et al. (2016) similarly find that communication moves group choices toward rationality more reliably than the aggregation of individual judgments alone, though for stochastic-dominance violations rather than the gambler's fallacy. There, communication

is fully interacted, so the implied treatment-specific coefficients reproduce those estimated separately in columns (1)–(3).

[22]Within the voting laboratory, Grosser and Seebauer (2016) report a slight tendency of the gambler's fallacy: completely uninformed votes favour the colour opposite to the previous round's true colour, at rates of 0.549 and 0.567 across their two experiments, in each case significantly above one half but modest in magnitude. Our analysis identifies this fallacy across specifications on our panel data, and in a stronger form: it continues to affect voting even when voters receive an informative private signal. More broadly, the literature on the gambler's fallacy has largely documented it where decision makers have no case-specific information; our evidence suggests the heuristic can be strong enough to override information they do hold.

helps largely through "demonstrability"—a member who recognizes the correct logic can articulate it to others (Laughlin and Ellis, 1986; Laughlin et al., 2006). Our mechanism is different: communication is too restricted to transmit reasoning and instead transmits members' private information. Speaking more directly to the gambler's fallacy, however, Stöckl et al. (2015) find that neither communication nor joint group decision making reduces it. The contrast with our finding that communication helps may be related to the information structure: in their setting, members observe the same history and receive no private signal about the current realization, whereas in ours, members hold heterogeneous private signals about the current state, giving communication new information to transmit and thereby helping voters override the heuristic.

6.2 Revisiting evidence on strategic abstention

The second noticeable pattern is the relatively low abstention rate (44.5%) in our NC treatment. Placing this finding alongside the broader experimental record suggests that the basic question—whether poorly informed voters abstain when strategic considerations favor delegation—is less settled than the early experimental evidence might suggest.

Table 6 collects the relevant treatments in the literature. The swing voter's curse was first tested experimentally by Battaglini et al. (2010), whose design gave voters either a perfect signal or none at all. In their non-partisan treatments, in which abstention is the unique equilibrium prediction, uninformed voters abstained 91% of the time when the prior was 1/2 but only 73% when it was 5/9. Abstention is high in both cases, but the drop is informative: abstention falls once voters have something directional to go on, even when that something is only a publicly shared prior.

Table 6: Abstention by poorly informed voters in laboratory experiments

Study, treatment	q_L	q_H	Info endogenous?	Equilibrium behavior	Abstention
Battaglini et al. (2010), non-partisan	No signal	1.00	No	Abstain	73.0–91.0%
Morton and Tyran (2011), ABS	0.65	0.90	No	Vote or abstain	87.9%
Großer and Seebauer (2016), voluntary	No signal	0.75	Mixed	Abstain	37.9–66.7%
Buechel and Mechtenberg (2019)	No signal	0.60	No	Abstain	63.0–75.0%
Elbittar et al. (2020), majority	No signal	0.67	Yes	Abstain	24.0–39.0%
This paper, NC	0.60	0.90	No	Vote or abstain	**44.5%**

Notes: All treatments have common interests. q_L and q_H denote the accuracy of low- and high-information voters' private signals; "No signal" denotes no private signal. Equilibrium behavior and abstention rates refer to low-information voters. Unless noted otherwise, treatments use a balanced prior and no pre-vote communication. Ranges correspond to $\pi = 1/2$ and $\pi = 5/9$ in Battaglini et al. (2010) (91.0% and 73.0% abstention); four communication-network structures in Buechel and Mechtenberg (2019); five voluntary-voting treatments in Grosser and Seebauer (2016), of which only Voluntary–Strangers–ExSignals assigns low-information status exogenously; and three- and seven-member committees in Elbittar et al. (2020). We omit the other treatments in Morton and Tyran (2011), where smaller information asymmetries make the equilibrium with non-expert abstention non-Pareto-optimal and thus not comparable to those reported here.

Morton and Tyran (2011) extend the evidence for strategic abstention to environments with somewhat informed rather than completely uninformed voters. In their ABS treatment ($q_L = 0.65$, $q_H = 0.9$), non-experts abstain at a high rate of 87.9%. Our NC treatment ($q_L = 0.6$, $q_H = 0.9$), however, yields an abstention rate of only 44.5%. The two treatments are close in design, and although some differences remain, none obviously accounts for a gap of this size.[23] Note that in both Morton and Tyran (2011) and our NC game, equilibria exist in which non-experts abstain as well as equilibria in which they vote their signals. Equilibrium analysis therefore does not uniquely pin down the abstention rate; how much non-experts abstain is ultimately an empirical question. By contrast, in Battaglini et al. (2010) and the other studies in the table, uninformed voters have no private signal, and abstention is uniquely predicted by the equilibria they focus on.

Reluctance to abstain, however, is not confined to our experiment. Substantial uninformed voting has also been documented in common-interest settings by Buechel and Mechtenberg (2019), Grosser and Seebauer (2016), and Elbittar et al. (2020). Although these papers focus on different questions—network structure in the first and endogenous costly information acquisition in the other two—all find that completely uninformed voters frequently vote rather than abstain. In all three, as in Battaglini et al. (2010), abstention is the unique equilibrium prediction. One possible explanation is that voters do not fully grasp the strategic logic of the swing voter's curse and fail to condition on pivotality (Niederle and Vespa, 2023). Consistent with this account, uninformed voting declines with experience in both Grosser and Seebauer (2016) and Buechel and Mechtenberg (2019), though it remains substantial throughout.[24] Our NC treatment shows the same pattern: non-expert abstention rises from 36.4% in the first ten rounds to 52.5% in the last ten ($p = 0.004$).[25] Elbittar et al. (2020) instead attribute uninformed voting to biased priors, classifying roughly 40% of subjects as biased types. The gambler's fallacy identified above in our NC treatment can thus be seen as one specific source of such biased priors.

Taken together, the experimental evidence suggests that strategic abstention by

[23] They differ mainly in three respects. First, our non-experts receive slightly less precise signals, which widens the quality gap and, if anything, should make abstention even more attractive. Second, we use 20 rather than 30 rounds, but non-experts in their ABS treatment already abstain 86.2% of the time over the first 20 rounds, so the additional rounds are unlikely to matter. Third, we use random rather than fixed matching. Fixed matching may aid coordination, but the gap is already large over the first ten rounds (81.7% versus 36.4%), so differential learning after those rounds is unlikely to be the main driver. We cannot rule out with the available data, however, that fixed matching helped their subjects coordinate on abstention within the first few rounds and thereby generated the early gap.

[24] Grosser and Seebauer (2016) also attribute uninformed voting to social pressure and negative reciprocity, two mechanisms specific to endogenous information that do not apply to our design.

[25] The non-expert silence rate in VC also rises from 33.9% to 48.4% ($p = 0.003$).

poorly informed voters is far from universal, and its prevalence varies substantially across settings. Further work is needed to identify when poorly informed voters strategically delegate, and to what extent limited abstention reflects imperfect pivotal reasoning, directional subjective priors, or equilibrium selection.

7 Conclusion

We examine how pre-vote communication affects collective decision-making in the context of the swing voter's curse. In our model, a limited communication stage does not change the set of equilibrium outcomes, so its effect depends on equilibrium selection. Based on behavioral conjectures, we hypothesize that communication reduces the efficiency of collective decisions; in the experiment, however, it increases efficiency. The "talk helps" result is particularly notable because our binary-message protocol was expected to leave less scope for communication to improve efficiency. The gain comes primarily from curbing irrational votes by non-experts, a channel we did not anticipate because our hypotheses all presume rational voting, and secondarily from helping groups follow the expert's message even when some non-experts fail to remain silent. At the same time, we find that "excess talk can hurt" within communication treatments: groups perform worse when everyone speaks than when non-experts remain silent, consistent with our message-dilution hypothesis.

Three behavior patterns deserve emphasis. First, absent communication, non-experts vote against their informative private signals in a non-negligible share of cases. We identify the gambler's fallacy as one source, and find that its influence on votes—and hence on group outcomes—is largely attenuated once communication is available. Second, non-experts rarely abstain after communication, even though abstaining would restore efficient delegation when everyone has spoken. The resulting dilution therefore carries into the voting stage and reduces efficiency. Third, we document strategic silence by non-expert voters in pre-vote communication, which occurs about as often as strategic abstention in the no-communication game. Although both behaviors are predicted by efficient equilibria, they are observed in fewer than half of cases in our experiment.

Our findings yield two implications for collective decision-making in practice. First, pre-vote communication can be valuable in attenuating individual biases. When cases arrive in sequence but are nearly independent of one another, such as proposal or promotion evaluations, poorly informed members may be misled by history, and voting alone aggregates those biases; our results suggest that even limited communication before voting can

help substantially, as it exposes members to information from others and thus drives their votes away from history-based heuristics and toward information about the current case.

Second, who speaks matters for how much communication helps. The dilution effect captures an inefficiency in pre-vote communication: when information quality varies widely across members, efficiency can be higher if less-informed participants remain silent. This effect is most relevant when information quality is hard to convey, typically under protocols that restrict communication to coarse assessments, as in straw polls and questionnaires with summary disclosure. The same tension may persist under richer communication when the precision of information is difficult to convey credibly—for example, when members care about appearing well informed (Ottaviani and Sørensen, 2006).[26] Norms that encourage less-informed members to refrain from speaking, or protocol designs that allow selective rather than universal participation in communication, may therefore improve collective decision-making.[27] The relevant institutional question is thus not simply whether groups should communicate before voting, but how communication should be structured when members differ in information quality.

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[26]Natural extensions of our setup include richer communication, reputational concerns, and uncertainty about members' own precision or relative expertise. Richer communication admits an equilibrium with full revelation of signals and precisions, though whether participants play such an equilibrium is an empirical question. Reputational concerns and uncertainty about expertise, by contrast, may enlarge the scope for dilution by making less-informed members even more likely to speak: silence can signal lower competence, while uncertainty may create scope for overconfidence. We leave them for future work.

[27]In a setting involving cognitive tasks rather than information aggregation, Charness et al. (2020) find that imposing a cost on messages improves team performance by reducing the quantity of communication while raising its quality. The context differs from ours, but the insight that silence can be valuable is similar.

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Appendix A Proofs

A.1 Notation and prerequisites

We explicitly formalize the game and assumptions for the VC game, as it is the most complex of the three games. The other two games are defined similarly.

Nature chooses a state $\omega \in \{A, B\}$, both being equally likely. Voter $i \in \{H, L_1, L_2\}$ receives a private signal $s^i \in \{A, B\}$. The expert's signal s^H matches the true state of the world with probability q_H, i.e., $q_H = \Pr\{s^H = A \mid \omega = A\} = \Pr\{s^H = B \mid \omega = B\}$. The non-experts' signals match the true state of the world with probability q_L, i.e., $q_L = \Pr\{s^i = A \mid \omega = A\} = \Pr\{s^i = B \mid \omega = B\}$ for $i \in \{L_1, L_2\}$, with $1 > q_H > q_L > 0.5$. Each voter knows these probabilities q_L and q_H, the number of each type of voter, and his or her own type. After receiving s^i, voter i chooses a message $m^i \in \{A, B, \varnothing\}$. Then he or she observes $m^{-i} = (x, y)$ where $x, y \in \{A, B, \varnothing\}$ are the messages chosen by the other two voters. Denote voter i's history until this point as $h^i \equiv \{s^i, m^i; m^{-i}\}$. After observing the history, he or she chooses a vote $v^i \in \{A, B, \varnothing\}$. In this game, voter i's *strategy* includes a communication strategy $m_i(s^i) : \{A, B\} \to \{A, B, \varnothing\}$ and a voting strategy $v_i(h^i) : \{A, B\} \times \{A, B, \varnothing\} \times \{A, B, \varnothing\}^2 \to \{A, B, \varnothing\}$. A *strategy profile* is $\sigma = (m, v)$, which includes a communication profile $m = (m_H, m_{L_1}, m_{L_2})$ and a voting profile $v = (v_H, v_{L_1}, v_{L_2})$. An equilibrium, denoted as $e = (\sigma; \mu)$, constitutes a perfect

Bayesian equilibrium (PBE), where $\mu = (\mu_H, \mu_{L_1}, \mu_{L_2})$ represents the belief system in this game.

The following assumptions are made regarding the strategies considered by voters, as introduced in the paper:

1. If not abstaining, voters vote responsively: If h and h' are such that $v_i(h) = A$ (or B) and all non-empty elements in h are opposite to the corresponding non-empty elements in h', then $v_i(h') = B$ (or A).

2. If not silent, voters send messages consistent with signals: for all i, if m_i is not identically equal to $\varnothing$, then $m_i(A) = A$ and $m_i(B) = B$.

3. The expert never stays silent or abstains: $m_H(s) \neq \varnothing$ and $v_H(h) \neq \varnothing$ for all s and h.

To facilitate the proof, we introduce some notation and prerequisites. As part of μ_i, voter i believes that the state is A with probability $\pi_i^e = \pi_i^e(h^i)$, dependent on history h^i and the communication strategy profile m^e in equilibrium e. When there is no ambiguity, we suppress the voter superscript and write h for a generic history. For convenience, define the option preferred by voter i's posterior in equilibrium as $\tilde{\pi}_i^e$, i.e.,

$$\tilde{\pi}_i^e = \begin{cases} A & \text{if } \pi_i^e > 0.5, \\ B & \text{if } \pi_i^e < 0.5, \\ s^i & \text{if } \pi_i^e = 0.5, \end{cases}$$

where the last case is defined for simplicity and is not crucial for the proof. Sometimes we drop the superscript to denote a generic variable $\tilde{\pi} \in \{A, B\}$. We say that voter i *votes sincerely* at history h if $v_i(h) = \tilde{\pi}_i^e(h)$, and *votes insincerely* at history h if $v_i(h) \in \{A, B\}$ and $v_i(h) \neq \tilde{\pi}_i^e(h)$.

Throughout the proof, for $X \in \{A, B\}$, write the probability $\Pr(\omega = X \mid \cdot)$ as $P(X \mid \cdot)$. Note that we have

$$P(A \mid s^H = A, s^{L_1} = s^{L_2} = B) = \frac{0.5 \cdot q_H \cdot (1 - q_L)^2}{0.5 \cdot q_H \cdot (1 - q_L)^2 + 0.5 \cdot (1 - q_H) \cdot q_L^2} > 0.5,$$

given the assumption $q_H > f(q_L)$ made in the paper.

For the NC game, voter i receives a private signal $s^i \in \{A, B\}$ and then chooses a vote $v^i \in \{A, B, \varnothing\}$. His or her strategy in this game is defined as $v_i(s^i) : \{A, B\} \to \{A, B, \varnothing\}$.

We say that voter i *votes for his or her signal* if $v_i(A) = A$ and $v_i(B) = B$, *votes against his or her signal* if $v_i(A) = B$ and $v_i(B) = A$, and *abstains* if $v_i(A) = v_i(B) = \varnothing$. For the FC game, the model will be the same as the VC game, except that in the communication stage, voters choose m^i from $\{A, B\}$ rather than $\{A, B, \varnothing\}$.

A.2 Proof of Proposition 1

Let $f(q_L) \equiv \frac{q_L^2}{1-2q_L(1-q_L)}$. When $q_H > f(q_L)$, *ED* is more efficient than *MD*, with an advantage that can approach 25 percentage points. When $q_H < f(q_L)$, *MD* is more efficient than *ED*, with an advantage that can approach 9.6 percentage points.

Proof: In an expert-decide (ED) outcome equilibrium, the probability that the group decision is correct is the probability that the expert is correct, i.e., $\text{Eff}_{ED} = q_H$. In a majority-decide (MD) outcome equilibrium, the probability of the group making correct decisions is equal to the probability that the majority signal is correct. We have

$$\text{Eff}_{MD} = \Pr(\text{majority signal is correct}) = 2q_H q_L(1 - q_L) + q_L^2$$

Therefore, the efficiency of the ED outcome is higher than the MD outcome by

$$\Delta = \text{Eff}_{ED} - \text{Eff}_{MD} = q_H - 2q_H q_L(1 - q_L) - q_L^2$$

which is greater than zero if and only if $q_H > \frac{q_L^2}{(1-2q_L(1-q_L))} \equiv f(q_L)$, and less than zero iff $q_H < f(q_L)$. Taking the derivatives of Δ gives us

$$\frac{\partial \Delta}{\partial q_H} = 1 - 2q_L(1 - q_L) > 0 \quad \forall q_L \in (0.5, 1)$$

$$\frac{\partial \Delta}{\partial q_L} = 4q_H q_L - 2q_H - 2q_L < 0 \quad \forall q_H, q_L \in (0.5, 1)$$

Thus, the largest efficiency gap in each of the two cases is:

$$\sup_{q_H > f(q_L);\, 1 > q_H > q_L > 0.5} |\Delta(q_H, q_L)| \to \Delta(1, 0.5) = 0.25$$

$$\sup_{q_L < q_H < f(q_L);\, q_L \in (0.5,1)} |\Delta(q_H, q_L)| \to \max_{q_L \in (0.5,1)} |\Delta(q_L, q_L)|$$

$$= \max_{q_L \in (0.5,1)} 3q_L^2 - 2q_L^3 - q_L = 3q_L^2 - 2q_L^3 - q_L|_{6q_L^2 - 6q_L + 1 = 0} = 3q_L^2 - 2q_L^3 - q_L|_{q_L \approx 0.79} \approx 0.096$$

A.3 Proof of Proposition 2

We prove parts (i)–(iii) in turn. The first statement of the proposition then follows immediately.

A.3.1 Part (i): The NC game

The proof for the NC game equilibria proceeds in two steps. First, Lemma 1 shows that, in equilibrium, voters either vote according to their signals or abstain. In other words, voting against one's signal can never occur in equilibrium. Second, we examine each of the four labeled equilibrium candidates and show that only two of them are equilibria.

Lemma 1 *In the NC game, in equilibrium,* $\forall i, s$*, if* $v_i(s) \neq \varnothing$*, then* $v_i(s) = s$*.*

Proof: Without loss of generality, assume $s^i = A$ throughout the proof. Consider distinct $i, j, k \in \{H, L_1, L_2\}$. We will show that $v^i = B$ is strictly worse in expected payoff than either choosing A or abstaining ($\varnothing$), for any strategies v_j and v_k.

In the NC game the history reduces to the private signal, $h^i = s^i$, so Assumption 1 requires that $v_i(A) = A$ imply $v_i(B) = B$, and that $v_i(A) = B$ imply $v_i(B) = A$. Applying the same condition at $h = B$ shows that $v_i(A) = \varnothing$ if and only if $v_i(B) = \varnothing$. Each voter's strategy is therefore one of exactly three: voting for one's signal, voting against one's signal, or abstaining. The enumeration below is thus exhaustive.

Given a strategy profile and signal A, whether voter i is better off taking a specific action rather than the other depends on situations where these two actions lead to different group decisions. We denote this situation as the pivotal event $Piv_i^{a_1,a_2}$, where $a_1, a_2 \in \{A, B, \varnothing\}$ represent the two actions being compared. It suffices to consider the expected payoff conditional on $Piv_i^{a_1,a_2}$, since in other scenarios, the difference between the two actions does not affect the outcome. Denote the votes of the other two players as $v^{-i} = (v^j, v^k)$.

- If both j and k choose to abstain, voting for B is worse for voter i than abstaining. *Proof:* In this case, v^i is pivotal. Voting for B determines the group decision as B and yields an expected utility of $P(B|s^i = A, Piv_i^{B,\varnothing}) = P(B|s^i = A) < 0.5$, while abstaining results in a tie with an expected utility of 0.5.

- If both j and k vote for their signals or both vote against them, voting for B is worse for voter i than abstaining. *Proof:* The pivotal event will be $v^{-i} = (A, B)$, which in equilibrium implies $s^j \neq s^k$. In this pivotal event, voting for B yields an expected

utility for voter i of $P(B|s^i = A, Piv_i^{B,\varnothing}) = P(B|s^i = A, s^j \neq s^k) = P(B|s^i = A) < 0.5$, while abstention results in an expected utility of 0.5.

- If voter j votes for his signal and voter k votes against his signal, voting for B is worse for voter i than abstaining. *Proof:* The pivotal event is represented as $v^{-i} = (A, B)$, which implies that $s^j = s^k$. In these pivotal events, voting for B yields an expected utility of $P(B|s^i = A, Piv_i^{B,\varnothing}) = P(B|s^i = A, s^j = s^k) = P(B|s^i = A) < 0.5$, while abstention results in an expected utility of 0.5.

- If j abstains and k votes for his signal, voting for B is worse for voter i than abstaining. *Proof:* The pivotal event is $v^{-i} = (\varnothing, A)$, which implies $s^k = A$. In this pivotal event, voting for B results in a tie with an expected utility of 0.5, while abstention leads to a group decision of A, which yields an expected payoff of $P(A|s^i = A, Piv_i^{B,\varnothing}) = P(A|s^i = A, s^k = A) > 0.5$.

- If j abstains and k votes against his signal, voting for B is worse for voter i than voting for A. *Proof:* If $v^k = A$, voter i voting for A leads to a group decision of A, while voting for B will result in a tie. If $v^k = B$, voting for A will lead to a tie, while voting for B will lead to a group decision of B. Note that given this profile, $v^k = A$ implies $s^k = B$ and $v^k = B$ implies $s^k = A$. Therefore, in expectation, voting for A increases the expected payoff compared to voting for B by $P(s^k = A|s^i = A)\left[0.5 - P(B|s^i = s^k = A)\right] - P(s^k = B|s^i = A)\left[0.5 - P(A|s^i = A, s^k = B)\right] > 0$.

Thus far, we have demonstrated that, given any voting profile of the other two players, voting against one's signal is strictly worse than either abstaining or voting for one's signal. Consequently, voters will never vote against their signals in equilibrium. *QED.*

Proof of part (i). By Lemma 1 and Assumption 3, there are four equilibrium candidates $v = (v_H(s^H), v_{L_1}(s^{L_1}), v_{L_2}(s^{L_2}))$ for examination: $(s^H, \varnothing, \varnothing)$, $(s^H, s^{L_1}, \varnothing)$, $(s^H, \varnothing, s^{L_2})$, and (s^H, s^{L_1}, s^{L_2}). We will assess each candidate by checking whether each voter would deviate from abstention to voting for his or her signal, or vice versa.

First, $(s^H, \varnothing, \varnothing)$ is an equilibrium. For L_1, his vote is pivotal when it is different from s^H. If $s^{L_1} = A$, then in pivotal events $(v^H = s^H = B)$, voting for A leads to a tie and gives an expected payoff of 0.5, while abstention leaves the group decision to be B and produces the expected payoff $P(B|s^{L_1} = A, s^H = B) > 0.5$. Therefore, L_1 can not gain by deviating to voting for his signal. Similarly, L_2 would not.

Second, $(s^H, s^{L_1}, \varnothing)$ is not an equilibrium. Following the same reasoning as in the previous paragraph, L_1 is better off abstaining rather than voting for his signal, given

that L_2 abstains and H votes for his signal. Therefore, L_1 would deviate by abstaining. Similarly, $(s^H, \varnothing, s^{L_2})$ is not an equilibrium.

Third, (s^H, s^{L_1}, s^{L_2}) is an equilibrium. Given this profile, v^{L_1} is pivotal only when the signals of the other two voters are opposing. In pivotal events, voting for s^{L_1} determines the group decision as s^{L_1} and yields an expected payoff of $P(A|s^{L_1} = A, s^H \neq s^{L_2}) = P(A|s^{L_1} = A) > 0.5$. Conversely, abstaining leads to a tie and provides an expected payoff of only 0.5. Therefore, L_1 would not deviate. Similarly, L_2 would not.

Thus, we have shown that the NC game has exactly two equilibria: $(s^H, \varnothing, \varnothing)$ and (s^H, s^{L_1}, s^{L_2}), which we refer to as *Hvote* and *Allvote*, respectively, in the proposition. It is evident that for all signal realizations $\boldsymbol{s}$, *Hvote* yields group decisions that always align with the expert's signal (expert-decide, or *ED* outcome), whereas *Allvote* yields group decisions that consistently reflect the majority signals (majority-decide, or *MD* outcome). *QED.*

A.3.2 Part (ii): The VC game

We begin by establishing several lemmas for the VC game to facilitate the proof of part (ii) of Proposition 2. Lemma 2 illustrates how voters update their beliefs about the state of the world after communication in equilibrium. Lemma 3 demonstrates that voters either abstain or vote sincerely in equilibrium, meaning they never vote insincerely. Following these lemmas, we will proceed to prove part (ii). Note that the communication strategy profile in an equilibrium e is denoted as $m^e = (m_H(s^H), m_{L_1}(s^{L_1}), m_{L_2}(s^{L_2}))$. Under the counting convention in the paper, profiles that differ only by relabeling L_1 and L_2 are counted as one profile. Thus, due to Assumptions 2 and 3, there are three potential equilibrium communication profiles up to relabeling: *Hspeak*, *H1Lspeak*, and *Allspeak*. For *H1Lspeak*, we take L_1 to be the speaking non-expert without loss of generality.

Lemma 2 *In the VC game, in any equilibrium e,*

(1) $\tilde{\pi}^e_H = s^H$ for all h;

(2) For $i \in \{L_1, L_2\}$, the following holds:

(i) if $m^e = (s^H, \varnothing, \varnothing)$, then $\tilde{\pi}^e_i = s^H$ for all h and i;

(ii) if $m^e = (s^H, s^{L_1}, \varnothing)$, then $\tilde{\pi}^e_{L_1} = s^H$ for all h, and

$$\tilde{\pi}^e_{L_2} = \begin{cases} s^H & \text{if } s^H = s^{L_1}, \\ s^{L_2} & \text{if } s^H \neq s^{L_1}. \end{cases}$$

The case in which L_2 speaks and L_1 stays silent is identical up to relabeling.

(iii) if $m^e = (s^H, s^{L_1}, s^{L_2})$, then

$$\tilde{\pi}_i^e = \begin{cases} s^H & \text{if } s^H = s^j, \\ s^i & \text{if } s^H \neq s^j, \end{cases}$$

where $j \in \{L_1, L_2\}$, $j \neq i$.

Proof: Without loss of generality, suppose $s^H = A$.

(1) In any equilibrium e, for any history h, we have $\pi_H^e \geqslant P(A|s^H = A, s^{L_1} = s^{L_2} = B) > 0.5$, where the equality holds if $s^{L_1} = s^{L_2} = B$. Thus, $\tilde{\pi}_H^e = A$.

(2) We now show non-experts' belief update in equilibrium for each communication profile:

(i) If $m^e = (s^H, \varnothing, \varnothing)$, then in equilibrium $m^H = A$ and voter $i \in \{L_1, L_2\}$ infers from $m^{-i} = (A, \varnothing)$ that $s^H = A$. For any s^i, we have $\pi_i^e = P(A|s^H = A, s^i) \geqslant P(A|s^H = A, s^i = B) > 0.5$. Thus, $\tilde{\pi}_i^e = s^H$ for all $i \in \{L_1, L_2\}$ and all h.

(ii) If $m^e = (s^H, s^{L_1}, \varnothing)$, following the same reasoning as in case (i), voter L_1 will have $\tilde{\pi}_{L_1}^e = s^H$ for all h. If $s^{L_1} = s^H$, then $m^{-L_2} = (A, A)$, voter L_2 infers that $s^H = s^{L_1} = A$, leading to

$$\pi_{L_2}^e = P(A|s^{L_2}, s^{L_1} = s^H = A) \geqslant P(A|s^{L_2} = B, s^{L_1} = s^H = A) > 0.5 \implies \tilde{\pi}_{L_2}^e = A.$$

If $s^{L_1} \neq s^H$, then $m^{-L_2} = (A, B)$, L_2 infers that $s^H \neq s^{L_1}$, thus

$$\pi_{L_2}^e = P(A|s^{L_2}, s^H \neq s^{L_1}) = P(A|s^{L_2}) \implies \tilde{\pi}_{L_2}^e = s^{L_2}.$$

The case in which L_2 speaks and L_1 stays silent is symmetric.

(iii) If $m^e = (s^H, s^{L_1}, s^{L_2})$, any voter $i \in \{L_1, L_2\}$ updates beliefs similarly to L_2 in case (ii).

QED.

Lemma 3 *In the VC game, in any equilibrium e for all i and h, if $v_i(h) \neq \varnothing$, then $v_i(h) = \tilde{\pi}_i^e$.*

Proof: From Assumption 3, $v_H(h) \neq \varnothing$ for all h. We first show that $v_H(h) \neq \tilde{\pi}_H^e$ will be worse than $v_H(h) = \tilde{\pi}_H^e$ for any e. Note that $\tilde{\pi}_H^e = s^H$ in any equilibrium e, according to Lemma 2. Suppose $s^H = A$; hence, $\tilde{\pi}_H^e = A$. In any equilibrium, if he votes for A, conditional on being pivotal, he gets the expected utility $P(A \mid h^H, Piv_H^{A,B}) \geqslant P(A \mid s^H = A, s^{L_1} = s^{L_2} = B) > 0.5$, greater than the expected payoff if he votes for B, $P(B \mid h^H, Piv_H^{A,B})$. Therefore, $v_H(h) = \tilde{\pi}_H^e$ for all h.

Then, without loss of generality, we show that $v_{L_1}(h) \neq \tilde{\pi}_{L_1}^e$ can never be optimal among non-abstaining votes against the others' voting profiles. There are three possible strategies of the others $(v_H(h), v_{L_2}(h))$. In the following, suppose $\tilde{\pi}_{L_1}^e = A$.

- If $(v_H, v_{L_2}) = (\tilde{\pi}_H^e, \varnothing)$, voting for B is worse for L_1 than abstention: Given this profile, $v^{L_1} = B$ is pivotal when $(v^H, v^{L_2}) = (A, \varnothing)$ which in equilibrium implies that $s^H = A$, as we have shown $v_H(h) = \tilde{\pi}_H^e$ for all h. Conditional on this pivotal event, L_1 gets an expected payoff of 0.5 if he votes for B, while if he abstains, he gets a group decision A and the expected payoff $P(A \mid h^{L_1}, Piv_{L_1}^{B,\varnothing}) = P(A \mid h^{L_1}, s^H = A) > 0.5$ given $\tilde{\pi}_{L_1}^e = A$. Therefore, voting against $\tilde{\pi}_{L_1}^e$ is worse than abstention.

- If $(v_H, v_{L_2}) = (\tilde{\pi}_H^e, \tilde{\pi}_{L_2}^e)$, voting for B is worse for L_1 than abstention: Given this profile, $v^{L_1} = B$ is pivotal when $(v^H, v^{L_2}) = (A, B)$ or (B, A), which in equilibrium implies that $\tilde{\pi}_H^e \neq \tilde{\pi}_{L_2}^e$, which implies $s^H \neq s^{L_2}$ by Lemma 2. Hence, conditional on the pivotal event, L_1 gets 0.5 if he abstains, while if he votes for B, he gets $P(B \mid h^{L_1}, Piv_{L_1}^{B,\varnothing}) = P(B \mid h^{L_1}, s^H \neq s^{L_2}) = P(B \mid h^{L_1}) < 0.5$ as $\tilde{\pi}_{L_1}^e = A$. Hence, voting against $\tilde{\pi}_{L_1}^e$ is worse than abstention.

- If $v_H = \tilde{\pi}_H^e$ but L_2 votes insincerely, voting B is worse for L_1 than abstention: Given the profile, $v^{L_1} = B$ is pivotal when $(v^H, v^{L_2}) = (A, B)$ or (B, A), which in equilibrium implies that $\tilde{\pi}_H^e = \tilde{\pi}_{L_2}^e$ but does not provide more information regarding the state of the world for L_1, according to Lemma 2. Hence, conditional on the pivotal event, L_1 gets 0.5 if he abstains, while if he votes for B, he gets $P(B \mid h^{L_1}, Piv_{L_1}^{B,\varnothing}) = P(B \mid h^{L_1}, \tilde{\pi}_H^e = \tilde{\pi}_{L_2}^e) = P(B \mid h^{L_1}) < 0.5$ as $\tilde{\pi}_{L_1}^e = A$. Hence, voting against $\tilde{\pi}_{L_1}^e$ is worse than abstention.

Therefore, for $i \in \{L_1, L_2\}$, if $v_i(h) \neq \varnothing$, then $v_i(h) \neq \tilde{\pi}_i^e$ can never occur in equilibrium. *QED.*

Proof of part (ii). Up to relabeling the two non-experts, define the following three com-

munication profiles:

$$m^1 = (s^H, \varnothing, \varnothing), \quad m^2 = (s^H, s^{L_1}, \varnothing), \quad m^3 = (s^H, s^{L_1}, s^{L_2}), \tag{1}$$

where in m^2 we take L_1 to be the speaking non-expert without loss of generality. Define the following voting profiles:

$$v^1 = (\tilde{\pi}^e_H, \varnothing, \varnothing), \quad v^2 = (\tilde{\pi}^e_H, \tilde{\pi}^e_{L_1}, \varnothing), \quad v^3 = (\tilde{\pi}^e_H, \varnothing, \tilde{\pi}^e_{L_2}), \quad v^4 = (\tilde{\pi}^e_H, \tilde{\pi}^e_{L_1}, \tilde{\pi}^e_{L_2}). \tag{2}$$

For m^1 and m^3, v^2 and v^3 are the same profile up to relabeling L_1 and L_2. For m^2, they are distinct because L_1 is the speaking non-expert and L_2 is the silent non-expert. Hence, up to relabeling, there are ten candidate profiles: (m^1, v^1), (m^1, v^2), (m^1, v^4); (m^2, v^1), (m^2, v^2), (m^2, v^3), (m^2, v^4); and (m^3, v^1), (m^3, v^2), (m^3, v^4). We show that eight of these are equilibria: the three *Hspeak* equilibria (m^1, v^1), (m^1, v^2), and (m^1, v^4); the three *H1Lspeak* equilibria (m^2, v^1), (m^2, v^2), and (m^2, v^4); *Allspeak–Hvote* (m^3, v^1); and *Allspeak–Allvote* (m^3, v^4). The remaining two candidate profiles, (m^2, v^3) and (m^3, v^2), are not equilibria.

By Lemmas 2 and 3, along with the assumptions, the profiles above are the only potential equilibrium profiles up to relabeling. We examine potential profitable deviations in each stage for each candidate. By Assumptions 2 and 3, deviations in the communication stage are restricted to non-experts, who may choose either silence or to send a message that aligns with their signals. As a result, any unilateral deviation in communication will lead the non-deviating players to see a different number of non-empty messages than in equilibrium, reaching an off-equilibrium information set. For any equilibrium candidate profile, assume that the associated belief system is such that any unilateral deviation in communication does not alter any voter's posterior belief on the state of the world, which consequently does not change votes or payoffs. Hence, no profitable deviation exists in the communication stage. Therefore, to determine whether a profile is an equilibrium, we only need to examine whether the non-experts would deviate in the voting stage from abstention to voting sincerely, or vice versa.

1. The three *Hspeak* equilibria: (m^1, v^1), (m^1, v^2), and (m^1, v^4).

Proof: By Lemma 2, in any equilibrium with $m^1 = (s^H, \varnothing, \varnothing)$ and for all $\boldsymbol{s} = \{s^i\}_{\forall i}$, after the communication stage, we have $\tilde{\pi}^e_H = \tilde{\pi}^e_{L_1} = \tilde{\pi}^e_{L_2} = s^H$. Given any of the voting profiles v^1, v^2, or v^4, voter $i \in \{L_1, L_2\}$ is indifferent between abstention and voting sincerely,

as his vote will never be pivotal. Specifically, his vote, if cast, will always align with the non-empty votes from the others. Therefore, neither non-expert will deviate, and all three profiles are equilibria. The omitted profile (m^1, v^3) is identical to (m^1, v^2) up to relabeling L_1 and L_2. It is immediate that these equilibria yield the group decision $D = v^H = \tilde{\pi}^e_H = s^H$ for all signal realizations $\boldsymbol{s}$, which is *expert-decide (ED)* by definition.

2. The three *H1Lspeak* equilibria: (m^2, v^1), (m^2, v^2), and (m^2, v^4); (m^2, v^3) is not an equilibrium.

Proof: By Lemma 2, in any equilibrium e with $m^2 = (s^H, s^{L_1}, \varnothing)$, after the communication stage, we have $\tilde{\pi}^e_H = \tilde{\pi}^e_{L_1} = s^H$ for all h, while $\tilde{\pi}^e_{L_2} \neq s^H$ for some h.

Consider $((s^H, s^{L_1}, \varnothing), (\tilde{\pi}^e_H, \varnothing, \varnothing))$. Voter L_1 would not deviate to voting sincerely, since his vote always aligns with v^H and does not change the group decision. For voter L_2, if $\tilde{\pi}^e_{L_2} = A$, his vote becomes pivotal only if $v^{L_2} \neq v^H$, which implies $s^H = B$. In such a scenario, if L_2 deviates to voting sincerely, it results in a tie with an expected payoff of 0.5. However, by abstaining, the group decision remains B with a higher expected payoff of $P(B \mid h^{L_2}, s^H = B) > 0.5$, so L_2 would not deviate. Thus, (m^2, v^1) is an equilibrium.

Now consider $((s^H, s^{L_1}, \varnothing), (\tilde{\pi}^e_H, \tilde{\pi}^e_{L_1}, \varnothing))$. Neither L_1 nor L_2 would deviate here, since their unilateral deviations between voting and abstaining would not alter the group decision. Thus, (m^2, v^2) is an equilibrium.

In the case of $((s^H, s^{L_1}, \varnothing), (\tilde{\pi}^e_H, \varnothing, \tilde{\pi}^e_{L_2}))$, it has been shown in the earlier proof that, given the fixed profile of the other players, L_2 benefits more from abstaining than voting. Thus, L_2 would deviate, and (m^2, v^3) is not an equilibrium.

Finally, consider $((s^H, s^{L_1}, \varnothing), (\tilde{\pi}^e_H, \tilde{\pi}^e_{L_1}, \tilde{\pi}^e_{L_2}))$. Suppose $\tilde{\pi}^e_{L_1} = A$, which means $\tilde{\pi}^e_H = s^H = A$. In this scenario, L_1's vote is pivotal only if $v^H \neq v^{L_2}$, implying $\tilde{\pi}^e_{L_2} = s^{L_2} = B$. If this pivotal event occurs, L_1 can secure the group decision as A by voting, yielding a payoff of $P(A \mid h^{L_1}, Piv^{A,\varnothing}_{L_1}) > 0.5$. If L_1 abstains, the group decision remains random, resulting in a payoff of 0.5. Therefore, L_1 would not deviate by abstention. Since $v^H = v^{L_1}$ always holds in equilibrium, L_2 would also not deviate as his vote is never pivotal. Thus, (m^2, v^4) is an equilibrium.

To summarize, (m^2, v^1), (m^2, v^2), and (m^2, v^4) are the three *H1Lspeak* equilibria up to relabeling. In each, for any signal realization $\boldsymbol{s}$, the group decision D is always secured to be s^H by either $v^H = s^H$ and two empty votes, or $v^H = v^{L_1} = s^H$, which already constitutes a majority vote. By definition, these equilibria produce *ED* outcomes.

3. The two *Allspeak* equilibria: (m^3, v^1) and (m^3, v^4); (m^3, v^2) is not an equilibrium.

Proof: By Lemma 2, in an equilibrium e with $m^3 = (s^H, s^{L_1}, s^{L_2})$, after the communication stage, any two voters' posteriors will prefer the same option given some h, but prefer different options given other h values. For each $i \in \{L_1, L_2\}$, there is a positive probability that his vote is pivotal in the equilibrium path. Suppose $\tilde{\pi}^e_{L_1} = A$.

Consider $\big((s^H, s^{L_1}, s^{L_2}), (\tilde{\pi}^e_H, \varnothing, \varnothing)\big)$. Given the profile, $v^{L_1} = A$ will only be pivotal if $v^{-L_1} = (B, \varnothing)$, which implies $\tilde{\pi}^e_H = s^H = B$. In pivotal events, L_1's vote sincerely for A will result in a tie (payoff of 0.5), while abstention results in the group decision B, which gives L_1 an expected payoff of $P(B \mid h^{L_1}, Piv^{A,\varnothing}_{L_1}) = P(B \mid h^{L_1}, s^H = B) > 0.5$. Hence, L_1 would not deviate by voting. Similarly, L_2 would not deviate. Hence, (m^3, v^1) is an equilibrium. It is immediate that in the equilibrium path, $D = v^H = s^H$ for all $\boldsymbol{s}$, which is an *ED-outcome*.

Now consider $\big((s^H, s^{L_1}, s^{L_2}), (\tilde{\pi}^e_H, \tilde{\pi}^e_{L_1}, \tilde{\pi}^e_{L_2})\big)$. Given the profile, $v^{L_1} = A$ will be pivotal if $v^{-L_1} = (A, B)$, which implies $\tilde{\pi}^e_H \neq \tilde{\pi}^e_{L_2}$. By Lemma 2, pivotal events indicate that $s^H \neq s^{L_2}$. Thus, if L_1 votes sincerely, he secures the group decision A and an expected payoff of $P(A \mid h^{L_1}, Piv^{A,\varnothing}_{L_1}) = P(A \mid h^{L_1}, s^H \neq s^{L_2}) = P(A \mid h^{L_1}) > 0.5$. Conversely, if L_1 abstains, he receives an expected payoff of 0.5 from the tie. Therefore, L_1 would not deviate by abstaining. Similarly, L_2 would not deviate. Hence, (m^3, v^4) is an equilibrium.

In this equilibrium, we have $\tilde{\pi}^e_i \neq s^i$ if $m^j = m^k \notin \{s^i, \varnothing\}$ and $i \neq H$, while $\tilde{\pi}^e_i = s^i$ otherwise. For signal realizations $\boldsymbol{s}$ where $\tilde{\pi}^e_i = s^i$ for all i, the voting stage results in $v^i = s^i$ for all i, making the group decision D determined by the majority of signals. Conversely, for signal realizations $\boldsymbol{s}$ such that $\tilde{\pi}^e_{L_2} \neq s^{L_2}$, we must have $m^H = m^{L_1} \notin \{s^{L_2}, \varnothing\}$, which implies $s^H = s^{L_1}$. In this latter case, by Lemma 2, we have $\tilde{\pi}^e_H = \tilde{\pi}^e_{L_1}$; thus, $v^H = v^{L_1} = s^H$ always holds, leading to $D = s^H = s^{L_1}$, which corresponds to the majority signal. Hence, this equilibrium is an *MD-outcome*.

Finally, consider $\big((s^H, s^{L_1}, s^{L_2}), (\tilde{\pi}^e_H, \tilde{\pi}^e_{L_1}, \varnothing)\big)$, which represents the one-non-expert-votes profile up to relabeling. As shown above, given the voting profile of the other players, the abstaining non-expert would be better off voting sincerely rather than abstaining. Therefore, (m^3, v^2) is not an equilibrium; the profile (m^3, v^3) is identical up to relabeling. *QED.*

A.3.3 Part (iii): The FC game

Proof of part (iii). In the FC game, the communication strategy is restricted to be *Allspeak*, i.e., $m^3 = (s^H, s^{L_1}, s^{L_2})$ as defined in equation (1). The *Allspeak* cases of Lemmas 2 and 3 also hold for the FC game. Following exactly the same reasoning as in the proof of part (ii) above, the profiles (m^3, v^1), i.e., *Allspeak-Hvote*, and (m^3, v^4), i.e., *Allspeak-Allvote*, are equilibria and have *ED* and *MD* outcomes, respectively. The other two voting profiles, (m^3, v^2) and (m^3, v^3), are not equilibria by the same argument as in part (ii). *QED.*

Appendix B Experiment instructions

B.1 Main instructions

Welcome to the experiment. We kindly request that you read the instructions carefully. This printed copy provides the same instructions that you will see on the screen.

During the experiment, it is important that you do not talk to other participants. If you have a question, please raise your hand and wait for an experimenter to come to you. You can then pose your question in private.

You will receive a show-up fee of 3.5 Euros and earnings on top of that, which depend on the decisions made by you and other participants in the lab. During the experiment, your earnings will be denoted in points. We will pay you afterward according to the exchange rate: 40 points = 1 Euro.

All decisions made in the experiment will remain anonymous. Participants will not be aware of each other's identities.

The experiment consists of 20 rounds, all following the same procedure. In the end, 2 rounds will be randomly selected for payment.

The Game

In each round, you will play a game in a group of three. There are two boxes: a red box referred to as RED, and a blue box referred to as BLUE. At the beginning of each round, a prize is randomly placed by the computer in one of the two boxes with equal probability. None of you knows where the prize is placed. The task of your group is to choose a box: if the prize is located in that box, each person of your group earns the prize; otherwise, you earn nothing. Each group member can vote for a box, and the group decision will be determined by majority rule. The detailed game rule will be explained

below.

The procedure of each round is as follows: (1) the prize is placed, (2) group members receive information, (3) group members send messages [Only in FC and VC treatments], (4) group members vote, (5) results are shown.

(1) The prize is placed

The computer randomly determines the placement of the prize, with an equal likelihood for either of the two boxes. In other words, there is a 50% probability that the prize is placed in BLUE and a 50% probability that the prize is placed in RED. This step is completed automatically by the computer before you see the first screen of a round. Neither you nor other members of your group know which box the prize is placed in.

(2) Group members receive information

Each group member receives information about where the prize is placed. The information is not 100% reliable but is always more likely to be correct than incorrect. Members will receive information with different probabilities of correctness:

- One member of the group receives information that is correct with 90% probability.
- The other two members receive information that is correct with 60% probability.

Your information is private, meaning that the other members do not know your information. Also, the group members do not necessarily receive the same information. For example, suppose you receive information that is correct with 60% probability, another member of your group also receives information that is correct with 60% probability and the last member receives information that is correct with 90% probability. In this case it is possible that you receive the information suggesting RED while the other two members receive the information suggesting BLUE.

It is randomly decided at the beginning of each round who gets which type of information.

You will receive your information and its reliability in the following form (note that this is just an example): *"The information you receive: 'The prize is in RED,' which is correct with 60% probability."*

(3) Group members send messages [Only in FC and VC treatments]

Each member sends a message to the group. You will choose to send a message that

says either ‘My information suggests RED’, ‘My information suggests BLUE’, or nothing (that is, you do not send any message) [The ’nothing’ option show up only in the VC treatment].

After everyone has made this choice, all group members will learn all the messages.

(4) Group members vote

Each member will choose to either ‘vote for RED‘, ‘vote for BLUE’ or ‘abstain’ (that is, do not cast a vote).

(5) Results are shown

The group decision will be made by majority rule. That is, the box that receives more votes will be the group decision. If RED and BLUE receive the same number of votes, the group decision will be made by chance, with both possibilities being equally likely.

Your earnings are determined exclusively by the group decision and the location of the prize. If your group finds the prize, i.e., the voting outcome is RED (BLUE) and the prize is placed in RED (BLUE), each group member earns 120 points in this round. Otherwise, everyone earns no points.

After the voting stage, the round ends. You will be told the number of votes for each box, the group decision, and the box that contains the prize.

By double clicking HISTORY at the bottom of your pages you will get to see a table with all your results (your information, the messages, the votes, the group decision, the place of the prize) from the previous rounds.

In the beginning of every round, you will be randomly rematched with two new participants. Nobody will know which other participants are in their group in any round.

Notice (Repetition of some important points) - Every round you will be randomly assigned to a new group. - The placement of prize, members’ information and who gets which type of information will be reset every round. - Your earnings are determined exclusively by the group decision and the location of the prize. Hence, they do not depend on the message that you send. - The 2 rounds for payment will be randomly selected from the 20 rounds.

If you finish reading, please click ’Next’ on your screen to answer comprehension questions. Afterward, you will play one practice round before playing the formal rounds. This practice round will not be paid.

B.2 Post-experiment questionnaire

Demographic Information

1. Your age is: _____ years old

2. Your gender is:

   Female Male Non-binary Prefer not to say Other (please specify): _____

3. Your current educational status is:

   Bachelor student Master student PhD student Prefer not to say Other (please specify): _____

4. Your nationality is: _____

5. Any comments or feedback? (for example, whether and how clear, interesting, or difficult the experiment is; how you make decisions; or any other observations you would like to share.)

Hypothetical Question (Not Affecting Your Payment)

1. Suppose that in a group, one person has information suggesting BLUE with a 90% correctness probability, while the other two persons both have information suggesting RED with a 60% correctness probability. To maximize their chances of winning the prize, which box should this group choose?

   RED BLUE I do not know / It is hard to say

2. If you chose RED or BLUE in the question above, how confident are you about your answer? (from 1–5 points, where a higher number means that you are more confident) 1 2 3 4 5

Appendix C Additional tables and analysis

C.1 More on non-expert behavior

Table C1 reports detailed non-expert voting behavior after communication across different scenarios in the communication stage. Assume without loss of generality that the non-experts receive signal A. When non-experts receive $(A, \varnothing)$ from others—that is, the sole non-empty message confirming their own signal A—it is natural to vote for A, and indeed

Table C1: Detailed non-expert voting after communication (assume signal A)

Others' messages	Own message	VC				FC			
		vote A	vote B	abstain	N	vote A	vote B	abstain	N
$(A, \varnothing)$	Overall	**94.4%**	**3.0%**	**2.6%**	300				
	$\varnothing$	96.5%	0.0%	3.5%	170				
	A	95.1%	3.3%	1.6%	122				
	B	37.5%	62.5%	0.0%	8				
$(B, \varnothing)$	Overall	**12.3%**	**85.5%**	**2.2%**	227				
	$\varnothing$	0.0%	99.2%	0.8%	124				
	A	28.0%	68.0%	4.0%	100				
	B	0.0%	100.0%	0.0%	3				
(A, A)	Overall	**97.9%**	**0.4%**	**1.7%**	238	**97.5%**	**0.8%**	**1.8%**	399
	$\varnothing$	93.0%	1.4%	5.6%	71				
	A	100.0%	0.0%	0.0%	158	97.4%	0.8%	1.8%	391
	B	100.0%	0.0%	0.0%	9	100.0%	0.0%	0.0%	8
(B, B)	Overall	**3.4%**	**90.5%**	**6.1%**	179	**2.9%**	**89.1%**	**8.0%**	311
	$\varnothing$	1.9%	88.9%	9.3%	54				
	A	4.2%	90.8%	5.0%	119	3.0%	88.8%	8.2%	304
	B	0.0%	100.0%	0.0%	6	0.0%	100.0%	0.0%	7
(A, B)	Overall	**68.7%**	**22.6%**	**8.7%**	336	**58.3%**	**26.1%**	**15.6%**	570
	$\varnothing$	67.6%	12.0%	20.4%	108				
	A	71.8%	25.0%	3.2%	220	58.4%	25.8%	15.8%	562
	B	0.0%	100.0%	0.0%	8	50.0%	50.0%	0.0%	8

Notes: "N" denotes the number of observations. No observations exist for others' messages $(\varnothing, \varnothing)$.

they do so 94.4% of the time. When both other voters send the same non-empty message, whether (A, A) or (B, B) (confirming or contradicting the non-expert's own signal A), non-experts almost always vote with this majority message: 97.9% and 90.5% of the time, respectively, in VC, and 97.5% and 89.1%, respectively, in FC.

The other two scenarios are more subtle. When the others send $(B, \varnothing)$, non-experts vote for B 85.5% of the time, suggesting some degree of sophistication: they appear to recognize that the only other non-empty message comes from the expert, and therefore they should follow it. Specifically, when the non-expert's own message is A, he or she still votes for B 68% of the time, consistent with substantial play of the *H1Lspeak* equilibria reported in Table 4. Finally, when non-experts receive balanced messages (A, B), they vote for their own signal most of the time (68.7% in VC, 58.3% in FC). Two observations are notable. First, non-experts sometimes vote against their own signals (over 20% of the time in both communication games), possibly due to confusion or forms of bounded rationality, such as the gambler's fallacy identified in the NC treatment (see Section 6). Second, abstention is more frequent in FC than in VC, as mentioned in the main text,

suggesting that the mandatory communication protocol encouraged subjects to consider abstaining.

C.2 Other Additional Tables

Table C2: Efficiency by profile type and round difficulty

		Easy	Medium	Tricky	Overall
ED equilibria	NC	97.3%	90.7%	75.0%	91.0%
		222	225	84	531
	VC	96.5%	88.7%	78.0%	89.8%
		339	477	150	966
	FC	-	90.0%	100.0%	92.3%
		-	30	9	39
	Theoretical predictions	95.3%	90.0%	80%	
MD equilibria	NC	94.2%	94.5%	16.7%	80.2%
		156	165	72	393
	VC	92.0%	92.2%	15.0%	83.1%
		225	231	60	516
	FC	95.1%	93.3%	8.6%	86.3%
		546	489	105	1140
	Theoretical predictions	95.3%	90.0%	20.0%	
Non-equilibrium	NC: H1Lvote	91.8%	74.6%	44.8%	74.5%
		147	213	87	447
	NC: opposite votes	70.0%	64.0%	77.1%	68.5%
		180	258	105	543
	NC: others	100.0%	0.0%	-	50.0%
		3	3		6
	VC	100.0%	88.2%	58.0%	79.5%
		60	228	150	438
	FC	94.7%	86.2%	72.2%	81.8%
		57	414	270	741

Notes: The percentages refer to the empirical efficiencies of the corresponding groups. The numbers below the percentages indicate the number of observations—that is, the number of groups that played each profile and faced each level of round difficulty.

Table C3: Robustness of the gambler's-fallacy result

Dependent variable: 1 if the non-expert votes for B (against signal A), 0 otherwise.							
	NC					VC	FC
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Past truth A	0.411**	0.631***	0.648***	0.627***	0.666***	0.002	0.121
	(0.187)	(0.225)	(0.231)	(0.230)	(0.248)	(0.180)	(0.102)
Past signal A				0.009	-0.037	0.080	-0.170*
				(0.197)	(0.208)	(0.150)	(0.103)
Past abstain		-1.644***	-1.640***	-1.645***	-1.638***	-1.047	-1.487
		(0.586)	(0.606)	(0.587)	(0.605)	(1.195)	(1.350)
Past vote against signal		1.684***	1.677***	1.684***	1.677***	-0.212	0.789**
		(0.504)	(0.490)	(0.504)	(0.490)	(0.410)	(0.309)
Past win		-0.044	-0.050	-0.045	-0.050	-0.160	-0.053
		(0.264)	(0.277)	(0.264)	(0.278)	(0.218)	(0.112)
Past win × Past abstain		-1.193	-1.189	-1.192	-1.192	-0.175	0.529
		(1.069)	(1.121)	(1.076)	(1.128)	(1.253)	(1.347)
Past win × Past vote against signal		0.729	0.870	0.729	0.871	0.226	-0.422*
		(0.545)	(0.582)	(0.545)	(0.579)	(0.425)	(0.256)
Round fixed effects	No	No	Yes	No	Yes	Yes	Yes
Log likelihood	-515.22	-407.00	-396.22	-407.00	-396.21	-770.97	-755.98
# observations	1,216	1,216	1,216	1,216	1,216	1,216	1,216
Equality tests for Past truth A under columns (5)–(7): NC = VC: $p = 0.027$; NC = FC: $p = 0.038$; VC = FC: $p = 0.555$							

Notes: All columns report logit estimates. Column (1) includes only Past truth A. Column (2) reproduces the baseline NC specification in Table 5. Column (3) adds round fixed effects, column (4) adds Past signal A, and column (5) includes both. Columns (6) and (7) apply the specification in column (5) to VC and FC, respectively. Past signal A equals one if the subject's private signal in the previous round supports the same option as his or her current-round private signal, where the current signal is normalized as A. The equality tests at the bottom come from pooled fully interacted logit specifications using the controls in columns (5)–(7): all reported history regressors are allowed to have treatment-specific coefficients, and round fixed effects are treatment-specific. Standard errors, reported in parentheses, are adjusted for clustering at the matching-group level; the pooled equality tests cluster across all matching groups. *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

Table C4: Alternative non-expert voting behaviors in NC

	Vote with signal		Abstain	
	(1)	(2)	(3)	(4)
Past truth A	-0.378**	-0.343**	0.049	-0.039
	(0.169)	(0.170)	(0.136)	(0.149)
Past signal A		-0.122		0.191
		(0.134)		(0.168)
Past abstain	-2.200***	-2.247***	2.608***	2.663***
	(0.584)	(0.589)	(0.531)	(0.550)
Past vote against signal	-0.353	-0.345	-2.693**	-2.707**
	(0.312)	(0.324)	(1.128)	(1.147)
Past win	-0.182	-0.205	0.243	0.278
	(0.186)	(0.196)	(0.328)	(0.353)
Past win × Past abstain	-0.908	-0.882	0.912*	0.855*
	(0.619)	(0.630)	(0.516)	(0.519)
Past win × Past vote against signal	-0.546	-0.630	0.164	0.181
	(0.377)	(0.436)	(1.197)	(1.222)
Round fixed effects	No	Yes	No	Yes
Log likelihood	-664.60	-651.47	-560.84	-550.11
# observations	1,216	1,216	1,216	1,216

Notes: All columns report logit estimates for non-experts in NC. In columns (1) and (2), the dependent variable equals one if the non-expert votes with his or her private signal and zero otherwise. In columns (3) and (4), the dependent variable equals one if the non-expert abstains and zero otherwise. Columns (1) and (3) use the baseline history controls from Table 5. Columns (2) and (4) additionally control for Past signal A and include round fixed effects. Past signal A equals one if the subject's private signal in the previous round supports the same option as his or her current-round private signal, where the current signal is normalized as A. Standard errors, reported in parentheses, are adjusted for clustering at the matching-group level. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.